

SOFTWARE DESIGN DOCUMENT (SDD)

for the
Task Force Management System (TFMS)

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1 SCOPE

1.1 Identification

System Name	: Task Force Monitoring System
System Abbreviation	: TFMS
System Identification Number	: TFMS-CODEForce
CSCI Name	: Not applicable
CSCI Abbreviation	: Not applicable
CSCI Identification Number	: Not applicable

1.2 Overview Of the System

Purpose of the TFMS:

The Task Force Monitoring System (TFMS) is designed to replace the current manual, spreadsheet-based approach for managing lecturer task force assignments within the Faculty of Artificial Intelligence (FAI). The system aims to address critical operational challenges related to workload transparency, fairness in task distribution, and administrative efficiency, with below capabilities:

- **Role-Based Access Control:** Secure authentication and authorization mechanisms for different user roles (System Administrator, HOD, PSM, Lecturer, Faculty Management).
- **Data Management Services:** Creation, retrieval, update, and deletion operations for staff, task forces, and membership data.
- **Business Logic Processing:** Implementation of weightage calculation algorithms, fairness assessment rules, and workflow validation.
- **User Interface Services:** Provision of web-based interfaces tailored to each user role's requirements.
- **Reporting Generation Engine:** Generation of reports, dashboards, and visualizations for operational and strategic purposes.
- **Audit Services:** Comprehensive logging of all system activities for accountability.

Purpose of the CSCI TFMS:

Not applicable



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1.3 Overview of the Documents

This document describes the preliminary and detailed design of the TFMS. The design follows Object-Oriented Design principles and uses a Model-View-Template (MVT) architectural pattern with current supported framework. This SDD is organized as follows:

CHAPTER 1 SCOPE

Provides identification and overview of the TFMS system and documentation.

CHAPTER 2 REFERENCED DOCUMENTS

Lists all documents referenced in this SDD.

CHAPTER 3 PRELIMINARY DESIGN

Describes the architectural design, interfaces, and components of TFMS.

CHAPTER 4 DETAILED DESIGN

Provides detailed design of each component and class.

CHAPTER 5 CSCI DATA

Describes the data elements within TFMS.

CHAPTER 6 NOTES

Contains abbreviations and additional information.



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2 REFERENCED DOCUMENTS

The following documents, for which the exact is indicated, form part of the specification as far as everything specified hereafter is concerned. In the event of a discrepancy between the documents referred to here and the content of this specification, it is the content of the specification, which should be considered to be the background reference.

Copies of specifications, standards, drawings and publications requested by suppliers in contact with the specified supplying functions may be obtained by contacting the contracting agency or directly through the contracting office.

2.1 Government Documents

- | | | |
|-----|---------------|--|
| [1] | DoD-Std-2167A | Defense System Software Development |
| [2] | DoD-Std-2168 | Defense System Software Quality Program |
| [3] | MIL-Std-1521B | Technical Reviews and Audits |
| [4] | RDL-101 | General Thomson Software Development Reference System |
| [5] | RDL-105 | Conduct of Reviews and Audit Guide Thomson Software Development Reference System |
| [6] | RDL-318 | SDP Drafting Guidelines Thomson Software Development Reference System |

2.2 Non-Government Documents

This section consists of the contractual documents and non-contractual documents.

2.2.1 Contractual documents

- [7] User Requirements Specification for Taskforce Monitoring System (TFMS)
- [8] Software Development Plan (SDP) for the Taskforce Management System (TFMS) Version 1.0, Item No. SDP-TFMS-261225-V1-
- [9] Software Requirement Specifications (SRS) for the Task Force Management System (TFMS) Version 1.0, Item No. SRS-TFMS-140126-V1-



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2.2.2 Non-contractual document

[10] Lecture notes MANP1153 – SOFTWARE DESIGN, version 4.0.2024.28.10

[11] UML Reference Guide Object Management Group (2023), Unified Modeling Language

[12] Django Software Foundation. (2024). Django documentation. Retrieved from <https://docs.djangoproject.com/>

[13] Python Software Foundation. (2024). The Python Language Reference. Retrieved from <https://docs.python.org/3/reference/>

[14] Meta Platforms, Inc. (2024). *React documentation*. Retrieved from <https://react.dev/>

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3 PRELIMINARY DESIGN

3.1 External Interface Requirements

The Task Force Management System (TFMS) interacts with five main user types to satisfy all functional requirements. Figure 1 shows the interactions between TFMS and its external interfaces.

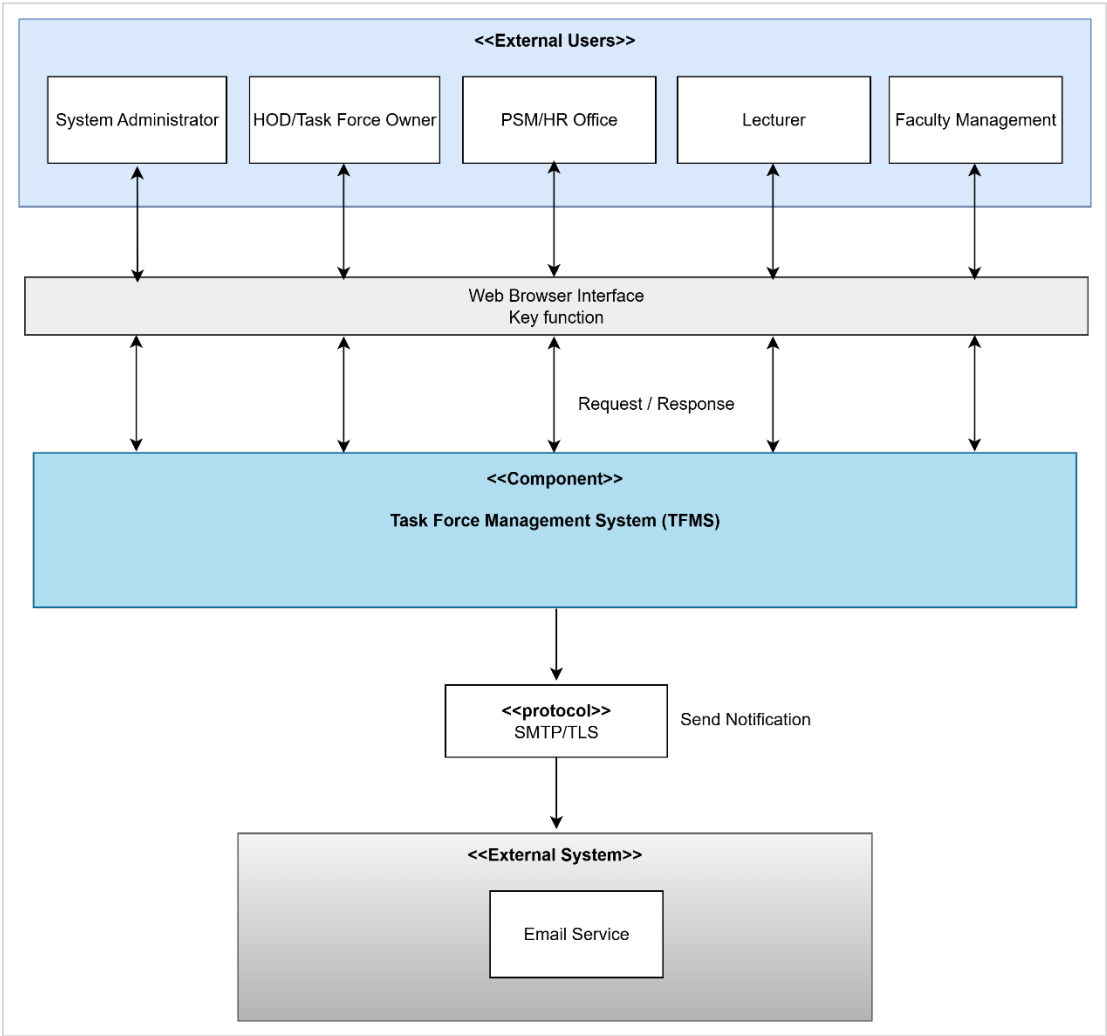


Figure 1: External Interface of TFMS

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3.1.1 External Users

i) System Administrator Interface

Interface Identification : **System Administrator**
Interface Type : **Web Browser Interface**

Description:

Manages master data (staff and task forces), system parameters, and user accounts.

Key Function:

Master data (staff and task forces), system parameters, and user accounts.

ii) HOD / Task Force Owner Interface

Interface Identification : **HOD / Task Force Owner**
Interface Type : **Web Browser Interface**

Description:

Proposes and updates task force membership, handles departmental allocations, and reviews staff workload.

Key Function:

Task force membership, departmental allocations, and staff workload.

iii) PSM / HR Officer Interface

Interface Identification : **PSM / HR Officer**
Interface Type : **Web Browser Interface**

Description:

Oversees faculty-level balancing, approval workflows, and reporting.

Key Function:

Faculty-level balancing, approval, and reporting.



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iv) Lecturers Interface

Interface Identification : **Lecturers**
 Interface Type : **Web Browser Interface**

Description:

View their task force portfolio, history, and workload status.

Key Function:

Task force portfolio, history, and workload status.

v) Faculty Management Interface

Interface Identification : **Faculty Management**
 Interface Type : **Web Browser Interface**

Description:

Accesses high-level dashboards and strategic summaries.

Key Function:

High-level dashboards and strategic summaries.

3.1.2 External System**i) Email Service – Outbound Notification Interface**

Interface Identification : Email Service (Outbound Notification)
Interface Type : SMTP/TLS Interface

Description:

This interface enables the TFMS to send system-generated email notifications to external users. It facilitates critical security features such as "Forgot Password" functionality via email-based token verification and ensures timely communication by delivering system alerts. These alerts are triggered by workflow events, including Task Force assignments, workload approval/rejection decisions, and account status updates (unlocking).

Key Function:

Send secure password reset links and account verification tokens. Notify HODs of workload submission status (Approved/Rejected).

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3.1.3 Internal API Interfaces

Not applicable yet

3.2 Architectural View

The TFMS architecture comprises of multitiers architectural model as illustrated in Figure 2. The TFMS package diagram is consisting of three tiers which are Presentation Layer, Application Layer and Data Layer.

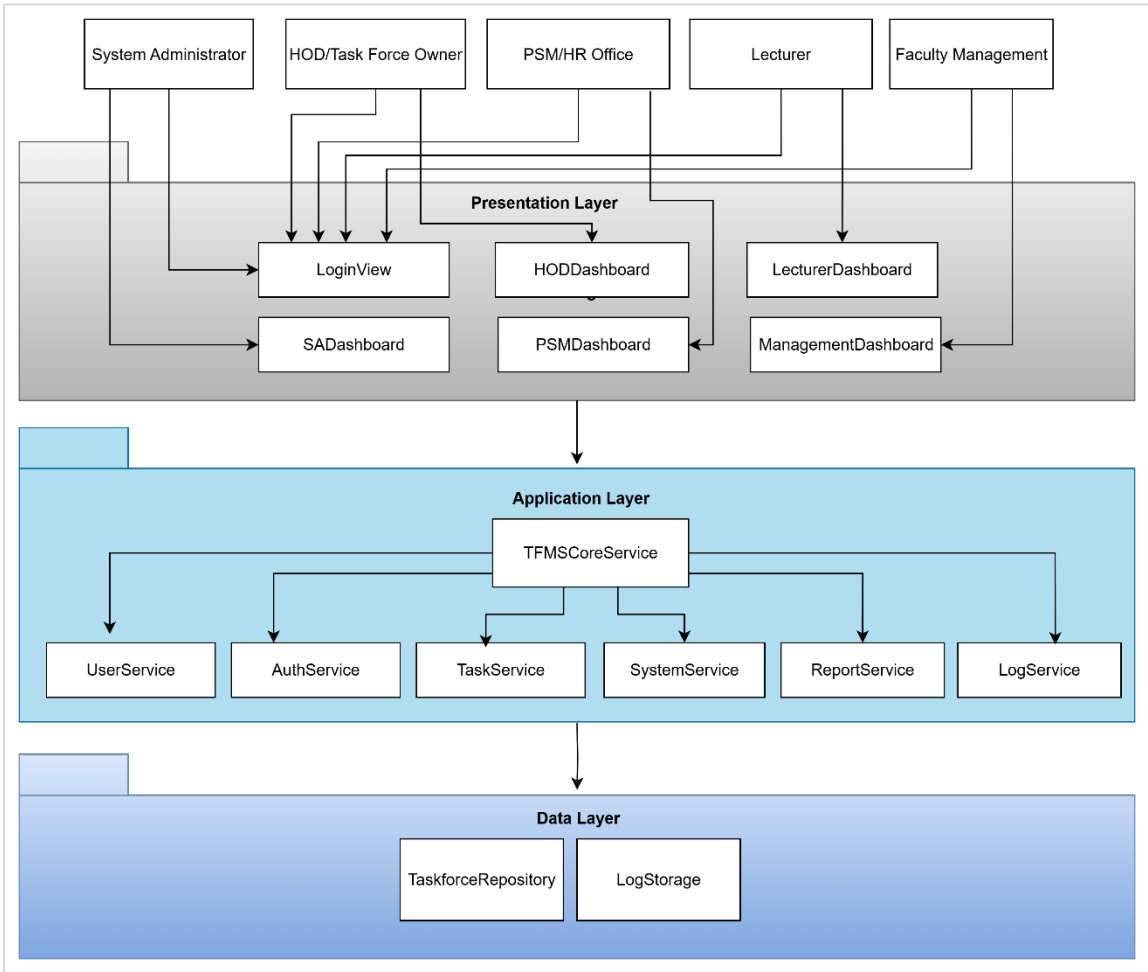


Figure 2: Top Level Architectural View of TFMS

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i) Presentation Layer

The presentation layer in TFMS is responsible for delivering role-specific user interfaces that facilitate secure access and intuitive interaction. It ensures that each actor engages with the system through tailored dashboards, promoting usability and separation of concerns from business logic.

This layer is composed of the following components within the UserInterface package:

- **LoginView:** Handles user authentication and entry into the system.
- **SADashboard:** Displays administrative controls and system-wide oversight for the System Administrator.
- **HODDashboard:** Provides task force management and workload allocation tools for Heads of Department.
- **PSMDashboard:** Facilitates approval workflows and reporting functions for PSM/HR Officers.
- **LecturerDashboard:** Presents workload status, task force history, and portfolio views for Lecturers.
- **ManagementDashboard:** Offers strategic summaries and faculty-level insights for Faculty Management.

ii) Application Layer

The application layer is responsible for coordinating system workflows and enforcing business processes in the TFMS. It acts as an intermediary between the presentation layer and the data layer, ensuring separation of concerns and modular system design.

At the application layer, the TFMSCoreService functions as a facade that coordinates core system work. The traceability matrix maps architectural flows by delegating responsibilities to specialized services such as UserService, TaskService, and ReportService. SystemService is introduced to manage system configuration concerns, aligning with the SystemConfig entity. This design avoids functional duplication by enforcing single-responsibility across services.

The application layer is composed of the following components within the CoreService package:

- **UserService:** Manages user accounts, roles, and access-related operations.
- **TaskService:** Handles task force creation, membership management, and workload-related operations.
- **ReportService:** Responsible for generating and exporting system reports.
- **AuthService:** Manages authentication and authorization processes.
- **SystemService:** Manages system configuration parameters and interacts with the SystemConfig entity.



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- **LogService:** a cross-cutting service to support traceability and compliance across multiple application services

iii) Data Layer

The data layer in TFMS is responsible for managing persistent storage and ensuring reliable access to system data. It supports the application layer by providing structured repositories and storage mechanisms that enforce data integrity, traceability, and modular access to core entities.

This layer is composed of the following components within the DataStore package:

- **TaskforceRepository:** Handles persistent storage and retrieval of task force definitions, memberships, and related configurations.
- **LogStorage:** Stores system-generated audit logs, enabling traceability and accountability across user actions and system events. AuditLog is mapped to a dedicated LogStorage to isolate audit concerns from transactional data, improving performance, security, and compliance.

Table 1 shows the traceability matrix dependency from top level architectural view presented using packages to component level.

Table 1: Traceability Matrix – Packages versus Components

No.	Layer	Packages	Component	Packages Stereotype
1.	Presentation	UserInterface	LoginView	<<component>>
			SADashboard	<<component>>
			HODDashboard	<<component>>
			PSMDashboard	<<component>>
			LecturerDashboard	<<component>>
			ManagementDashboard	<<component>>
2.	Application	CoreService	UserService	<<component>>
			AuthService	<<component>>
			TaskService	<<component>>
			ReportService	<<component>>
			SystemService	<<component>>
			LogService	<<component>>
3.	Data Logic	DataStore	TaskforceRepository	<<database>>
			LogStorage	<<database>>



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3.3 Static Organization

This section defines and describes the packages, components and classes and nodes for the TFMS.

3.3.1 Detailed Component Diagram

Figure 3 illustrates the component-level architecture of the TFMS. The diagram is organized into six primary packages representing key system layers and functionalities:

1. **UserInterface** – Contains components for user interaction, including different dashboards for PSM, SA, HOD, Lecturers, and Management, along with the LoginView.
2. **CoreService** – Encapsulates the core business logic of TFMS, such as UserService, TaskService, AuthService, ReportService, SystemService, and LogService. AccessControl ensures proper authorization across services.
3. **DataStore** – Manages persistent storage through components like TaskforceRepository and LogStorage.
4. **GlobalLib** – Provides reusable libraries and utilities, including MathLib, ErrorHandler, Caching, Utils, and ThirdPartyLib.
5. **CommService** – Handles system communication via APIService and EmailService.
6. **ExternalService** – Integrates external dependencies such as email systems, messaging queues, and third-party services.



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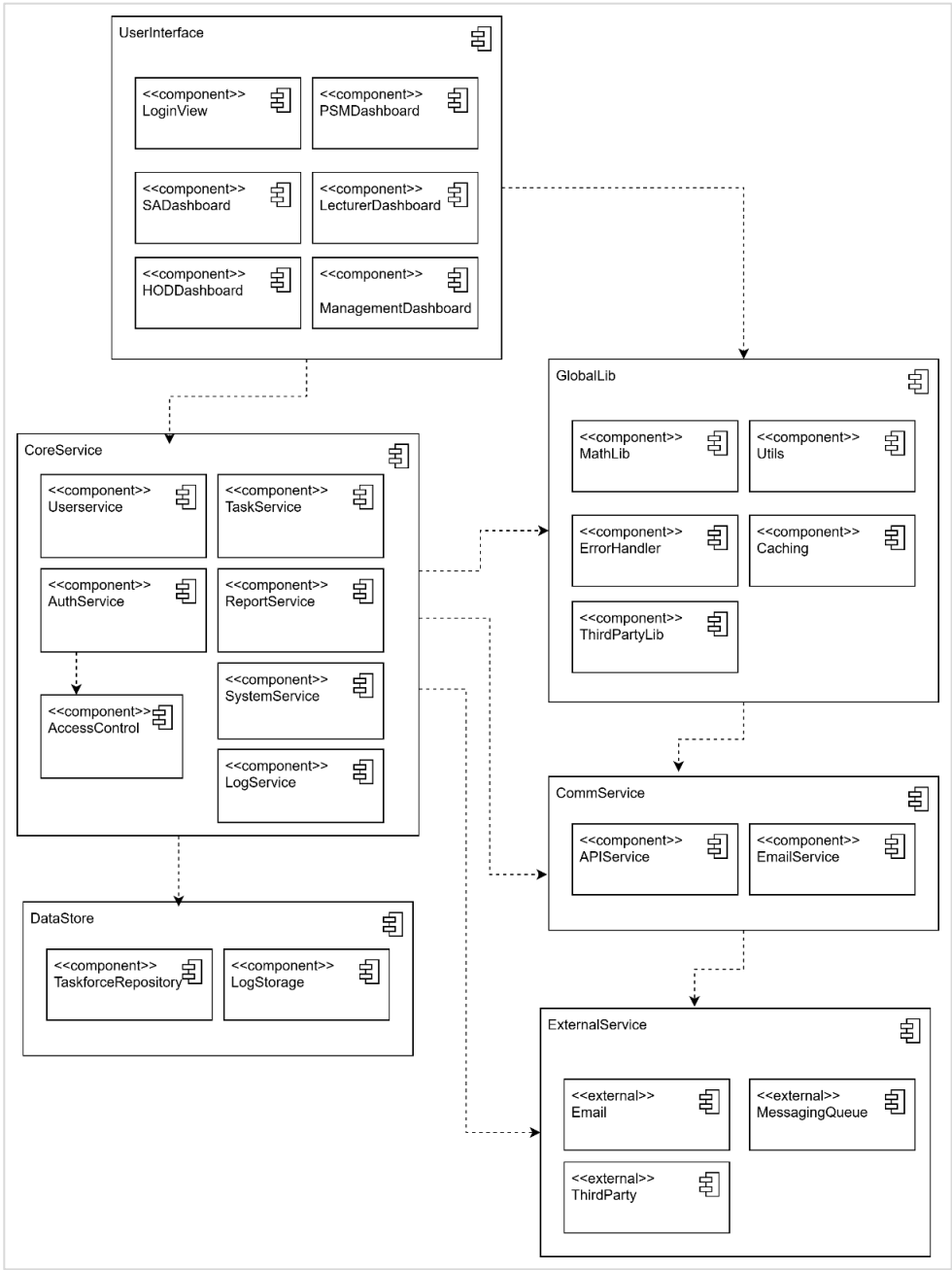


Figure 3: Component Diagram of TFMS

Table 2 presents the traceability matrix, which maps architectural components to their corresponding realizing classes. This mapping ensures alignment between the component-level design and the class-level model. Each component may be realized by one or more classes, reflecting the distribution of its functional responsibilities. Boundary components correspond to UI-related classes, executable components map to control and service classes, and data store components are associated with



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persistent entity classes. This traceability demonstrates alignment between architectural and detailed design artifacts of the TFMS.

Table 2: Traceability Matrix – Components versus Classes

No	Source Components	Component Stereotype	Classes	Classes Stereotype
1.	LoginView	Boundary	UI	Boundary
			User	Entity
2.	LecturerDashboard	Boundary	UI	Boundary
			User	Entity
3.	HODDashboard	Boundary	UI	Boundary
			User	Entity
4.	PSMDashboard	Boundary	UI	Boundary
			User	Entity
			ApprovalRequest	Entity
5.	ManagementDashboard	Boundary	UI	Boundary
			User	Entity
6.	TFMSScoreService	Control / Facade	TFMS	Control
7.	UserService	Executable	User	Entity
			Role	Entity
			UserRole	Entity
8.	TaskService	Executable	TaskForce	Entity
			TaskForceMembership	Entity
			Department	Entity
			WeightageRule	Entity
9.	AuthService	Executable	User	Entity
			Role	Entity
10.	ReportService	Executable	Report	Entity
11.	SystemService	Executable	SystemConfig	Entity
12.	LogService	Executable	AuditLog	Entity
13.	TaskforceRepository	Data Store	User	Entity
			Role	Entity
			UserRole	Entity
14.	LogStorage	Data Store	AuditLog	Entity
15.	SADashboard	Boundary	UI	Boundary
			User	Entity



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3.3.2 Detailed Class Diagram

The following Figure 4 below shows the detailed class diagram for TFMS. Each of the class has defined attributes, values and operations.

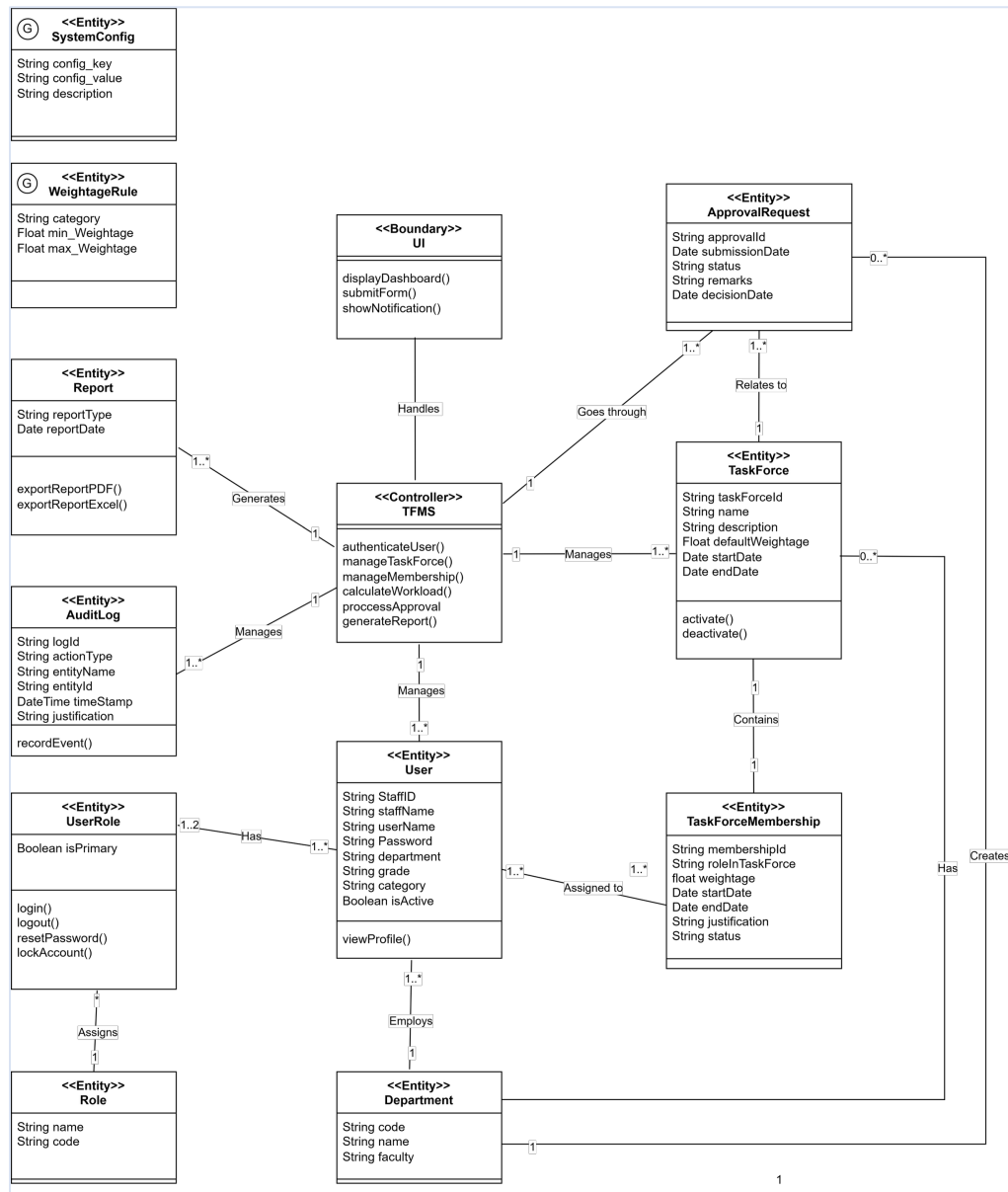


Figure 4: Detailed Class Diagram of TFMS

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3.3.3 Detailed Class Diagram of a Components

Figure 5 illustrates the class structure for the system's login and authentication process. This component consists of three primary classes that work together to verify user identity and grant access.

- **[TFMS-01-100-PY-CSU-1.1]-UI (Boundary):** This class functions as the user interface layer. It presents the login screen and captures user input. Key methods include *submitForm()* to send entered credentials for processing and *showNotification()* to display success or error messages to the user.
- **[TFMS-06-100-PY-CSU-6.1]-TFMS (Controller):** The TFMS class acts as the central controller for the system. It processes the logic behind user actions. Specifically, the *authenticateUser()* method receives data from the UI and verifies it against stored records. It also coordinates subsequent actions, such as dashboard retrieval.
- **[TFMS-13-100-PY-CSU-13.1]-User (Entity):** This entity class stores essential staff information required for the system's operation. It holds attributes such as *staffID*, *userName*, *password*, and *department*. The controller uses this data to validate credentials and determine the user's role and active status.



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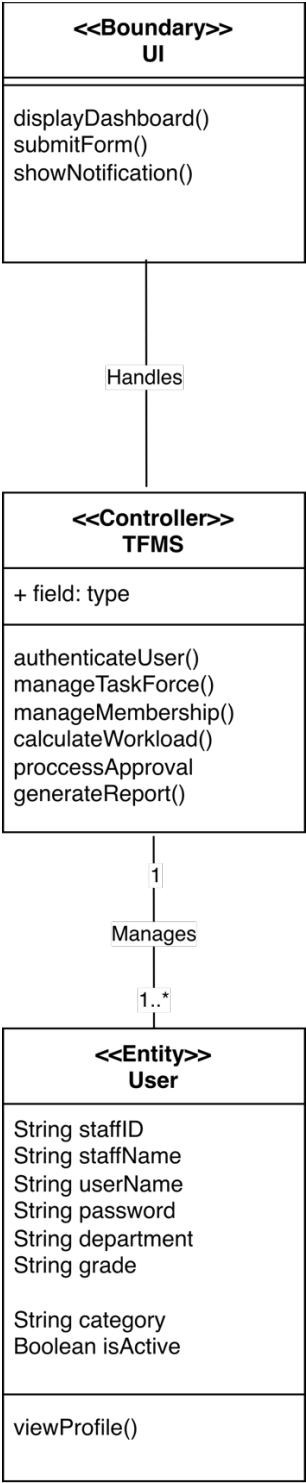


Figure 5: Detailed Class Diagram of Login Authentication Component



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Figure 6 depicts the classes involved in system administration, where the administrator manages core data structures and configurations. This component ensures the system's operational integrity through five key classes.

- **[TFMS-15-100-PY-CSU-15.1]-UI (Boundary):** This class represents the administration dashboard. It provides the interface for administrators to input data and view system status. The *submitForm()* method handles the creation and modification of records, while *showNotification()* provides immediate feedback on administrative actions.
- **[TFMS-06-100-PY-CSU-6.1]-TFMS (Controller):** This controller class executes the business logic for administrative tasks. The *manageTaskForce()* and *manageMembership()* methods handle the creation and updating of task forces and user accounts. It also utilizes *calculateWorkload()* to validate configurations against defined rules.
- **[TFMS-08-100-PY-CSU-8.1]-TaskForce (Entity):** This entity represents the working groups within the university. It stores critical details such as *taskForceId*, *name*, *description*, and *defaultWeightage*. The class includes methods like *activate()* and *deactivate()* to control the lifecycle of a task force.
- **[TFMS-15-100-PY-CSU-15.2]-User (Entity):** In the context of administration, this entity represents the staff records managed by the system. Administrators create and update these records, defining attributes like *staffID*, *role*, and *department*, which determine a user's access level and responsibilities.
- **[TFMS-08-100-PY-CSU-8.4]-WeightageRule (Entity):** This entity defines the constraints for workload distribution. It stores the *min_Weightage* and *max_Weightage* for different categories. These rules serve as the benchmark for the controller to validate assignments and prevent workload imbalances.
- **[TFMS-12-100-PY-CSU-12.1]-AuditLog (Entity):** This class ensures accountability by recording all critical system actions. It captures details such as *actionType*, *timeStamp*, and the acting user. The *recordEvent()* method persists these logs, creating a secure trail for security and compliance reviews.



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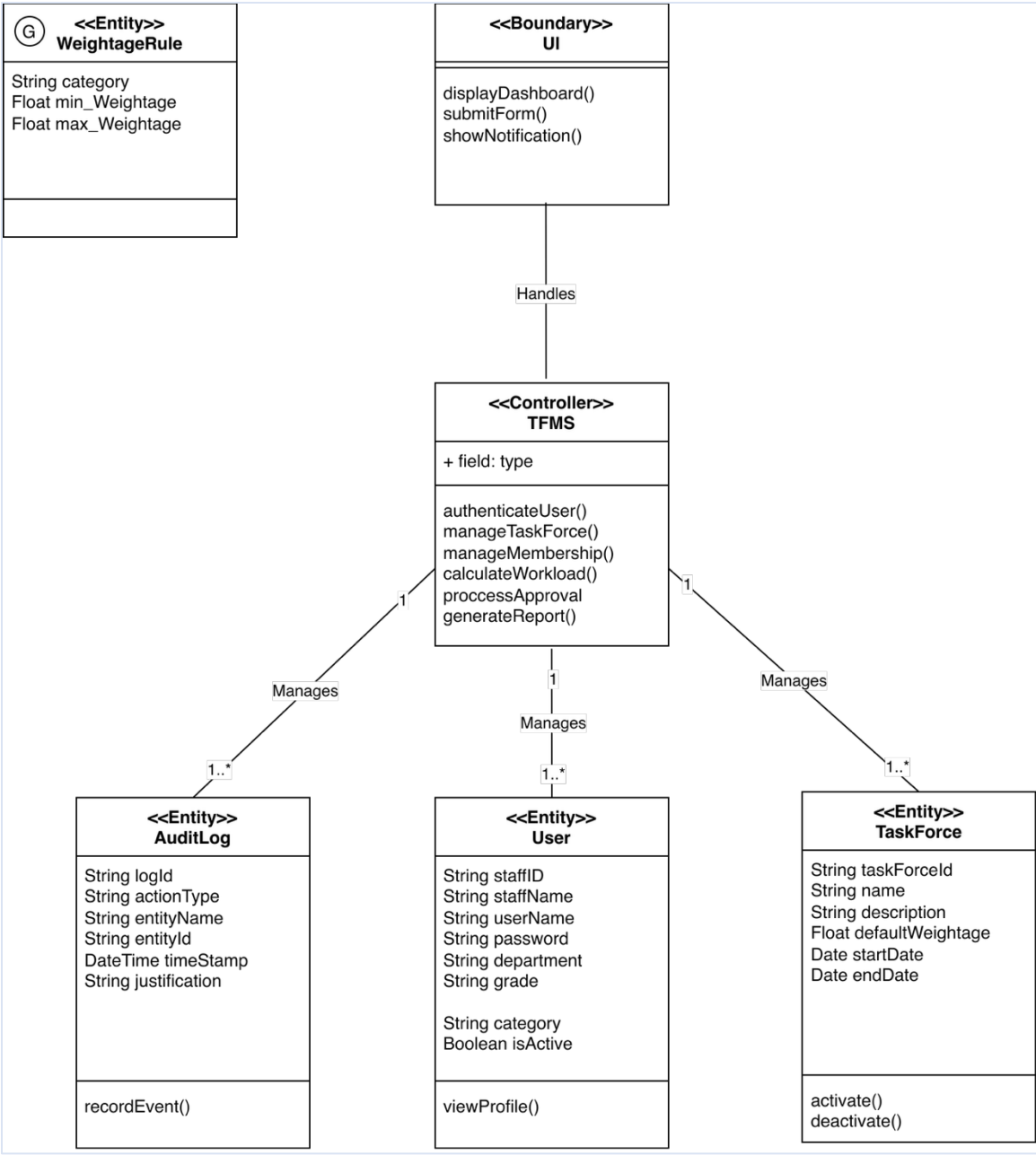


Figure 6: Detailed Class Diagram of System Administrator Component

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Figure 7 outlines the classes responsible for the Head of Department's (HOD) functionality, specifically focusing on task force management and membership assignment. This component coordinates user roles, task forces, and approval workflows.

- **[TFMS-03-100-PY-CSU-3.1]-UI (Boundary):** This class represents the HOD's interface for managing departmental task forces. It allows the HOD to select task forces, assign members, and view dashboard status updates via *displayDashboard()*. The *submitForm()* method captures membership assignments and submits them for approval.
- **[TFMS-06-100-PY-CSU-6.1]-TFMS (Controller):** The controller processes the HOD's logic. The *manageMembership()* method is central here, assigning potential members to task forces. It calls *calculateWorkload()* to ensure assignments comply with the thresholds defined in *WeightageRule*, triggering warnings if limits are exceeded.
- **[TFMS-08-100-PY-CSU-8.1]-TaskForce (Entity):** This entity represents the specific task force being managed. It serves as the container for the assignments. The HOD interacts with this entity to populate it with members and define its active duration using *startDate* and *endDate*.
- **[TFMS-03-100-PY-CSU-3.3]-User (Entity):** In this context, the User entity represents the pool of available staff (Lecturers) within the department. The controller filters these users based on their departments to present eligible candidates for assignment.
- **[TFMS-08-100-PY-CSU-8.4]-WeightageRule (Entity):** This entity provides the validation logic. When an HOD attempts to assign a member, the system checks the *min_Weightage* and *max_Weightage* to ensure the resulting workload is balanced.
- **[TFMS-04-100-PY-CSU-4.2]-ApprovalRequest (Entity):** This class tracks the submission of the assembled task force. Once the HOD finalizes assignments, an *ApprovalRequest* is created with a *submissionDate* and *status* (e.g., "Pending"), which is then routed to the PSM for review.
- **[TFMS-12-100-PY-CSU-12.1]-AuditLog (Entity):** This entity records the HOD's actions, such as "Member Assigned" or "Request Submitted," ensuring a historical record of all changes made to task force compositions.



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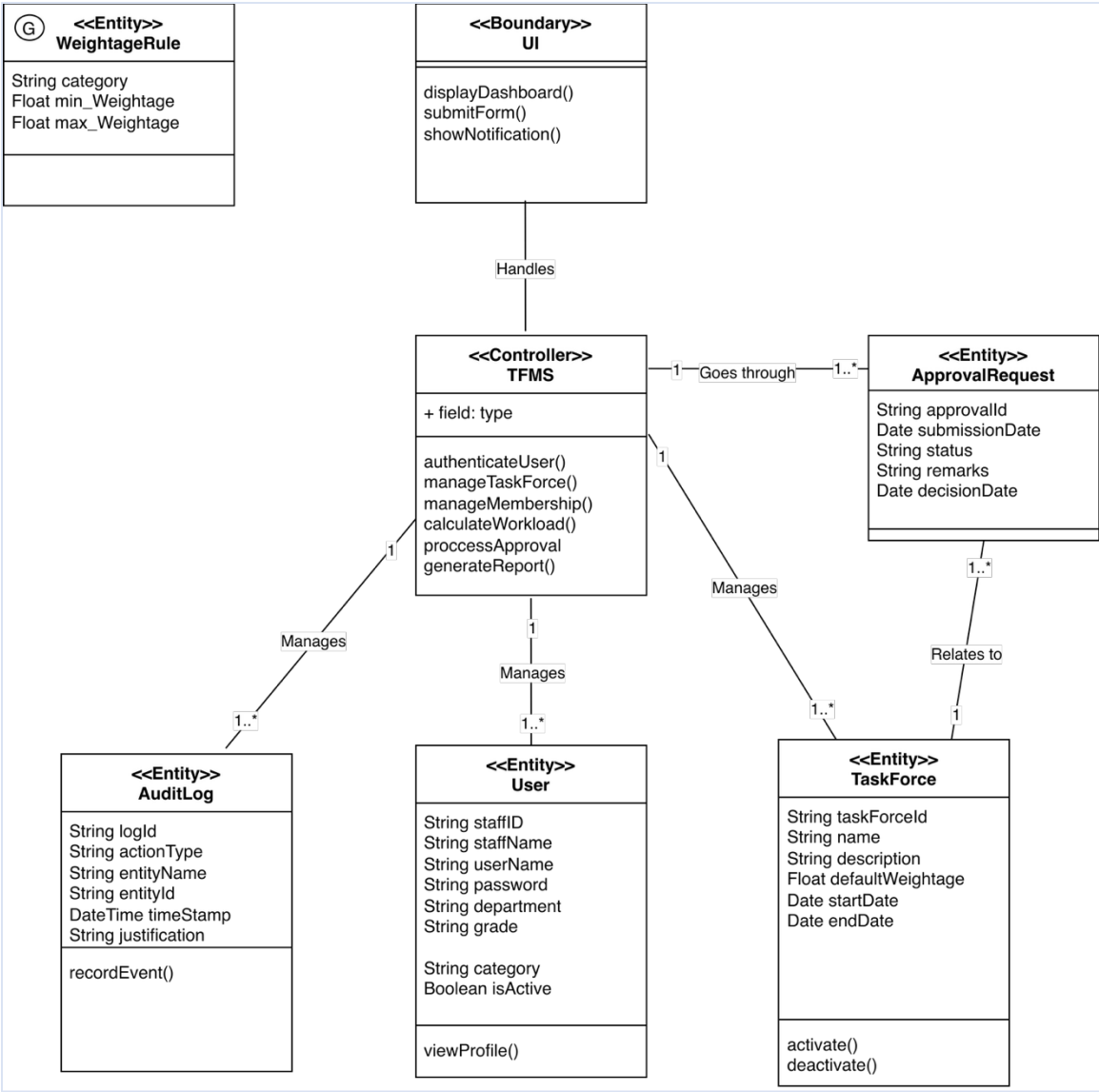


Figure 7: Detailed Class Diagram of HOD Component

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Figure 8 portrays the classes responsible for the Project Supervisor/Manager (PSM) operations, focusing primarily on the review and approval process of task force submissions. This component enforces governance and compliance through six key classes.

- **[TFMS-04-100-PY-CSU-4.1]-UI (Boundary):** This class serves as the approval interface for the PSM. It lists pending submissions and provides controls for review. The *submitForm()* method is used here to register the PSM's decision (Approve or Reject), while *showNotification()* confirms the action to the user.
- **[TFMS-06-100-PY-CSU-6.1]-TFMS (Controller):** The controller orchestrates the approval logic. The *processApproval()* method is the core function, retrieving pending requests, updating their status, and calculating the final workload impacts via *calculateWorkload()*.
- **[TFMS-04-100-PY-CSU-4.2]-ApprovalRequest (Entity):** This entity is central to the PSM's workflow. It represents the specific item being reviewed. It contains the *submissionDate*, current *status*, and allows the PSM to add *remarks*, which is critical for providing feedback on rejections.
- **[TFMS-08-100-PY-CSU-8.1]-TaskForce (Entity):** This entity represents the task force associated with the approval request. The PSM reviews its composition and details (like *name* and *defaultWeightage*) to ensure it meets institutional requirements before granting approval.
- **[TFMS-08-100-PY-CSU-8.4]-WeightageRule (Entity):** During the review process, this entity acts as the validation standard. The PSM uses the defined *min_Weightage* and *max_Weightage* to double-check that the proposed task force staffing levels are appropriate and equitable.
- **[TFMS-12-100-PY-CSU-12.1]-AuditLog (Entity):** This class maintains the integrity of the approval process. Every decision made by the PSM whether approval or rejection - is permanently recorded via *recordEvent()* with a timestamp and justification, ensuring a transparent audit trail.



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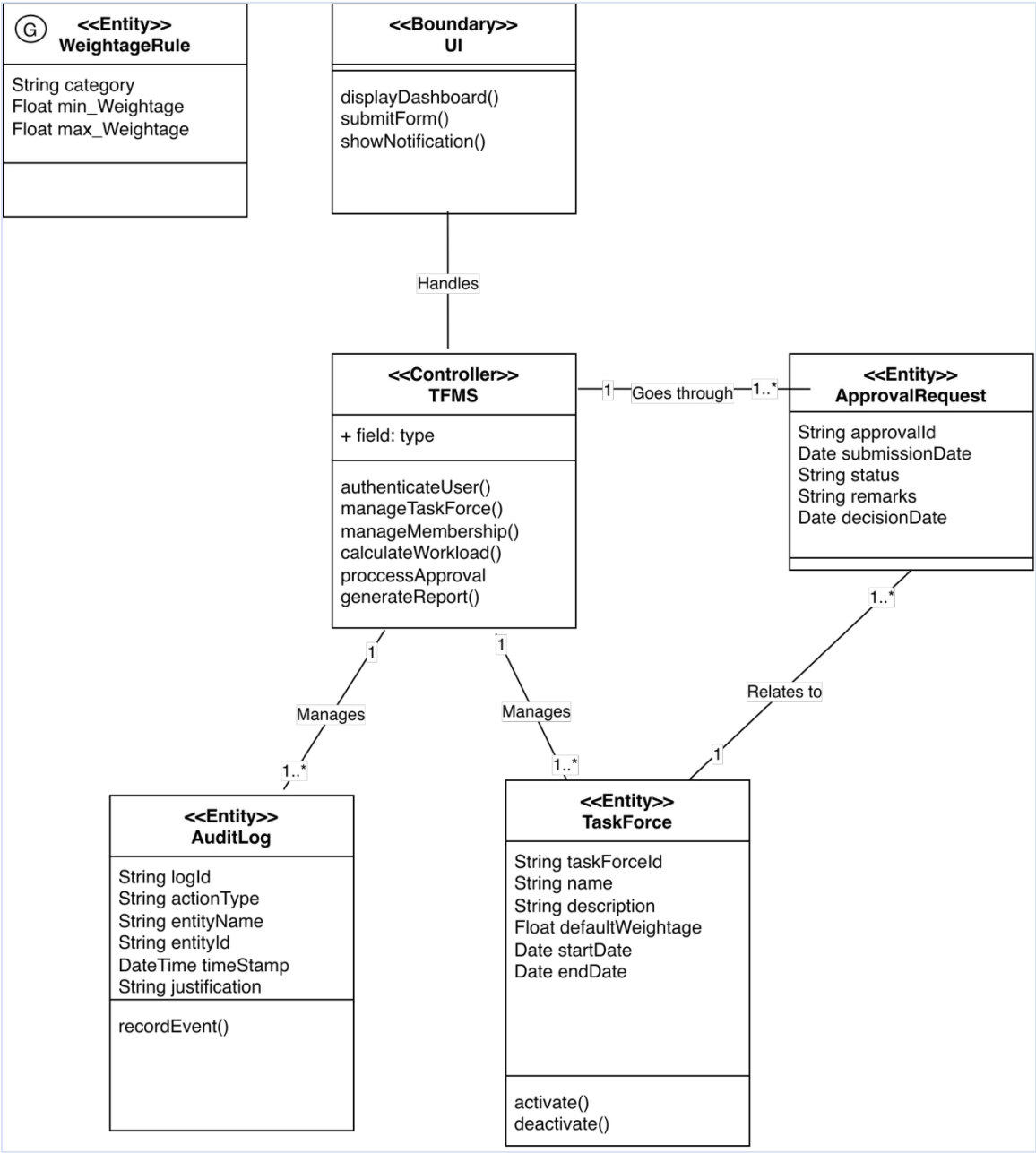


Figure 8: Detailed Class Diagram of PSM Component

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Figure 9 details the classes that facilitate the Lecturer's interactions with the system, focusing on portfolio access and reporting. This component empowers academic staff to view their assignments and generate personal records through six interconnected classes.

- **[TFMS-02-100-PY-CSU-2.1]-UI (Boundary):** This class serves as the Lecturer's personal dashboard. It gathers data to *displayDashboard()*, showing current task force assignments and workload status. It also allows the Lecturer to initiate report generation via the *submitForm()* method, and receives real-time updates through *showNotification()*.
- **[TFMS-06-100-PY-CSU-6.1]-TFMS (Controller):** The controller manages data retrieval for the lecturer. Its *generateReport()* method is the key function here, aggregating assignment data, calculating totals via *calculateWorkload()*, and preparing it for export.
- **[TFMS-08-100-PY-CSU-8.1]-TaskForce (Entity):** This entity represents the specific committees or groups the lecturer is assigned to. The controller queries these entities to populate the lecturer's portfolio, displaying details like *name*, *description*, and the assignment duration (*startDate* to *endDate*).
- **[TFMS-10-100-PY-CSU-10.1]-Report (Entity):** This specialized entity handles the output of the lecturer's data request. It encapsulates the *reportType* and *reportDate*, and provides specific methods such as *exportReportPDF()* and *exportReportExcel()* to format the aggregated data into downloadable files.
- **[TFMS-08-100-PY-CSU-8.4]-WeightageRule (Entity):** For the lecturer, this entity provides transparency regarding their workload. The controller uses the stored *min_Weightage* and *max_Weightage* values to visually indicate on the dashboard whether the lecturer's current load is balanced, under-utilized, or overloaded.
- **[TFMS-12-100-PY-CSU-12.1]-AuditLog (Entity):** This class tracks the lecturer's system usage, such as log-ins and report downloads. The *recordEvent()* method ensures that access to personal data and the generation of official records is logged for security purposes.



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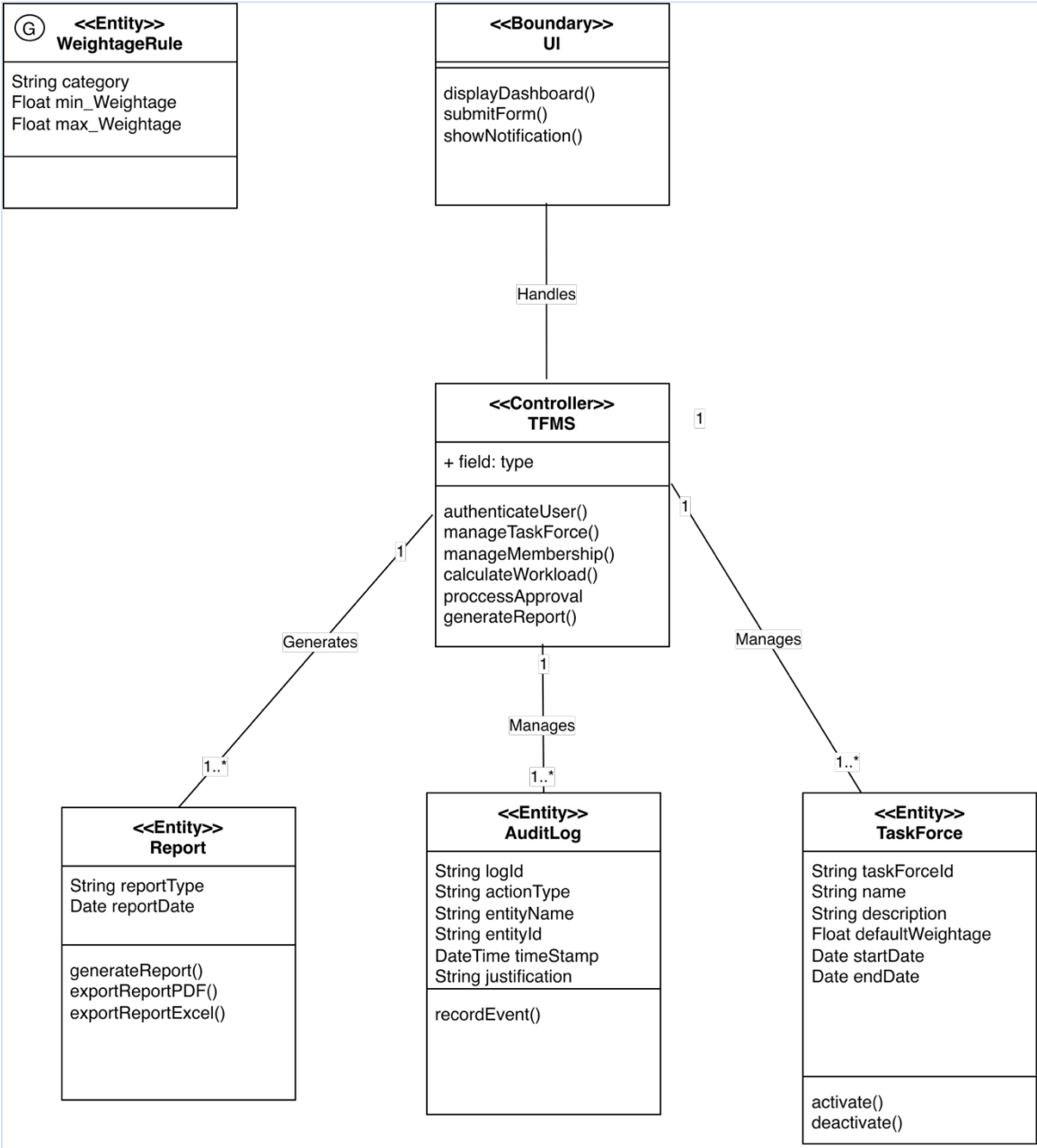


Figure 9: Detailed Class Diagram of Lecturer Component

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Figure 10 illustrates the classes supporting the Faculty Management (Dean's) component, which is designed for high-level oversight and strategic planning. This component relies on six classes to provide aggregated data and generate executive-level insights.

- **[TFMS-05-100-PY-CSU-5.1]-UI (Boundary):** This class represents the executive dashboard used by the Dean. It is optimized to *displayDashboard()* views that summarize departmental performance and system-wide statistics. It also serves as the interface for requesting complex reports via *submitForm()* and receiving generation status alerts through *showNotification()*.
- **[TFMS-06-100-PY-CSU-6.1]-TFMS (Controller):** The controller acts as the data aggregator. The *generateReport()* method is particularly robust here, capable of pulling data across all departments (task forces and users) to create comprehensive summaries. It also utilizes *calculateWorkload()* to provide macro-level views of faculty resource utilization.
- **[TFMS-08-100-PY-CSU-8.1]-TaskForce (Entity):** This entity represents the entire collection of active task forces across the faculty. The controller aggregates data from these entities, such as total counts, active status, and collective weightage to feed the management dashboard.
- **[TFMS-10-100-PY-CSU-10.1]-Report (Entity):** This entity generates formal executive documentation. It handles the creation of broad-scope documents defined by *reportType*, supporting critical decision-making by offering *exportReportPDF()* and *exportReportExcel()* capabilities for offline analysis and presentation.
- **[TFMS-08-100-PY-CSU-8.4]-WeightageRule (Entity):** In the management context, this entity provides the benchmarks for assessing faculty-wide workload distribution. It allows the Dean to verify if departments are adhering to institutional policies regarding workload balance (*min_Weightage* and *max_Weightage*).
- **[TFMS-12-100-PY-CSU-12.1]-AuditLog (Entity):** This class ensures governance by tracking access to sensitive high-level data. The *recordEvent()* method logs when executive reports are generated or when the Dean accesses global views, maintaining a secure record of administrative oversight.



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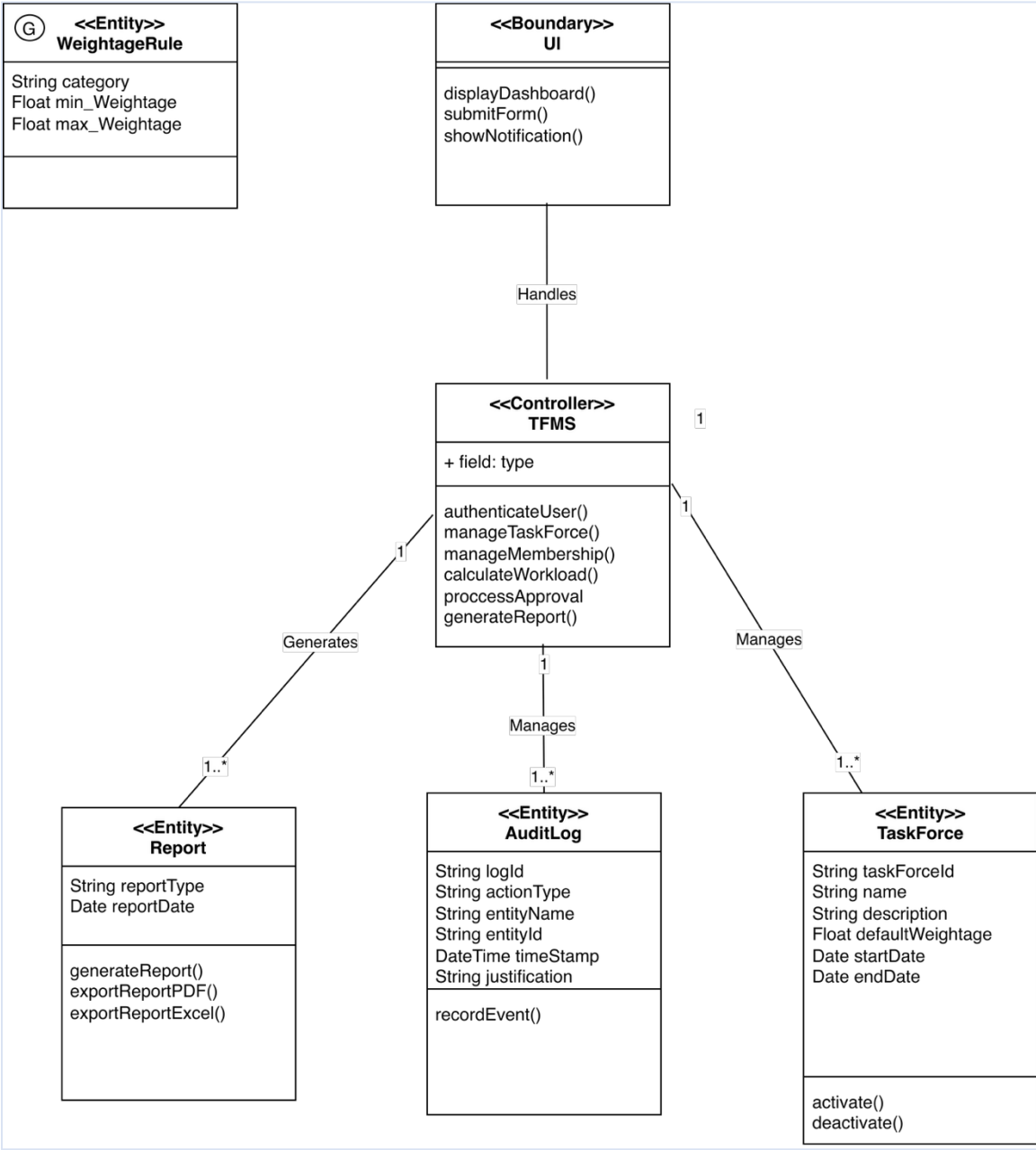


Figure 10: Detailed Class Diagram of Management Component

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The given identifications of the above components and classes are identified as the following Table 3:

Table 3: Traceability Matrix – Components and Classes Item Identification

No.	CSC Name	CSC Id	No	CSU Name	CSU Id
CSC-1	LoginView	TFMS-01-100-PY-CSC-1	CSU-1.1	UI	TFMS-01-100-PY-CSU-1.1
			CSU-1.2	User	TFMS-01-100-PY-CSU-1.2
CSC-2	LecturerDashboard	TFMS-02-100-PY-CSC-2	CSU-2.1	UI	TFMS-02-100-PY-CSU-2.1
			CSU-2.2	User	TFMS-02-100-PY-CSU-2.2
CSC-3	HODDashboard	TFMS-03-100-PY-CSC-3	CSU-3.1	UI	TFMS-03-100-PY-CSU-3.1
			CSU-3.2	User	TFMS-03-100-PY-CSU-3.2
CSC-4	PSMDashboard	TFMS-04-100-PY-CSC-4	CSU-4.1	UI	TFMS-04-100-PY-CSU-4.1
			CSU-4.2	User	TFMS-04-100-PY-CSU-4.2
			CSU-4.3	ApprovalRequest	TFMS-04-100-PY-CSU-4.2
CSC-5	ManagementDashboard	TFMS-05-100-PY-CSC-5	CSU-5.1	UI	TFMS-05-100-PY-CSU-5.1
			CSU-5.2	User	TFMS-05-100-PY-CSU-5.3
CSC-6	TFMSCoreService	TFMS-06-100-PY-CSC-6	CSU-6.1	TFMSController	TFMS-06-100-PY-CSU-6.1
CSC-7	UserService	TFMS-07-100-PY-CSC-7	CSU-7.1	User	TFMS-07-100-PY-CSU-7.1
			CSU-7.2	Role	TFMS-07-100-PY-CSU-7.2
			CSU-7.3	UserRole	TFMS-07-100-PY-CSU-7.3
CSC-8	TaskService	TFMS-08-100-PY-CSC-8	CSU-8.1	TaskForce	TFMS-08-100-PY-CSU-8.1
			CSU-8.2	TaskForceMembership	TFMS-08-100-PY-CSU-8.2
			CSU-8.3	Department	TFMS-08-100-PY-CSU-8.3
			CSU-8.4	WeightageRule	TFMS-08-100-PY-CSU-8.4



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No.	CSC Name	CSC Id	No	CSU Name	CSU Id
CSC-9	AuthService	TFMS-09-100-PY-CSC-9	CSU-9.1	User	TFMS-09-100-PY-CSU-9.1
			CSU-9.2	Role	TFMS-09-100-PY-CSU-9.2
CSC-10	ReportService	TFMS-10-100-PY-CSC-10	CSU-10.1	Report	TFMS-10-100-PY-CSU-10.1
CSC-11	SystemService	TFMS-11-100-PY-CSC-11	CSU-11.1	SystemConfig	TFMS-11-100-PY-CSU-11.1
CSC-12	LogService	TFMS-12-100-PY-CSC-12	CSU-12.1	AuditLog	TFMS-12-100-PY-CSU-12.1
CSC-13	TaskforceRepository	TFMS-13-100-PY-CSC-13	CSU-13.1	User	TFMS-13-100-PY-CSU-13.1
			CSU-13.2	Role	TFMS-13-100-PY-CSU-13.2
			CSU-13.3	UserRole	TFMS-13-100-PY-CSU-13.3
CSC-14	LogStorage	TFMS-14-100-PY-CSC-14	CSU-14.1	AuditLog	TFMS-14-100-PY-CSU-14.1
CSC-15	SADashboard	TFMS-15-100-PY-CSC-15	CSU-15.1	UI	TFMS-15-100-PY-CSU-15.1
			CSU-15.2	User	TFMS-15-100-PY-CSU-15.2

3.4 Deployment Architecture

Figure 11 illustrates the physical deployment environment of the TFMS as hosted on a free cloud platform.

Users access TFMS through User Devices using a standard web browser, which communicates over HTTPS with the hosted application. The system is deployed on a hosted web server (PythonAnywhere – Free Tier) and is accessible via: <https://junaed.pythonanywhere.com/>

The application runs on a Django (Python) web framework, which handles request routing, authentication, and service coordination. Within the Local Application (Python via Django), the core business logic is managed by TFMSCoreService, which coordinates several internal services including:

- AuthService and AccessControl for authentication and authorization
- UserService and TaskService for lecturer and task-force management
- ReportService, SystemService, and LogService for reporting, system operations, and logging

The application relies on commonly used Python libraries, including:

- Django Framework (web framework)



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- SQLite3 (lightweight embedded database)
- Google API Client Libraries (integration with Google services such as email or authentication, where applicable)
- Standard **Python Utilities** for logging, error handling, and data processing

Data persistence is handled using SQLite (MySQLite), deployed locally within the hosted environment, which stores task-force data and system logs.

TFMS also integrates with external services, including:

- Google APIs (explicitly used for external service integration)
- Email services for notification delivery
- Other external APIs as required



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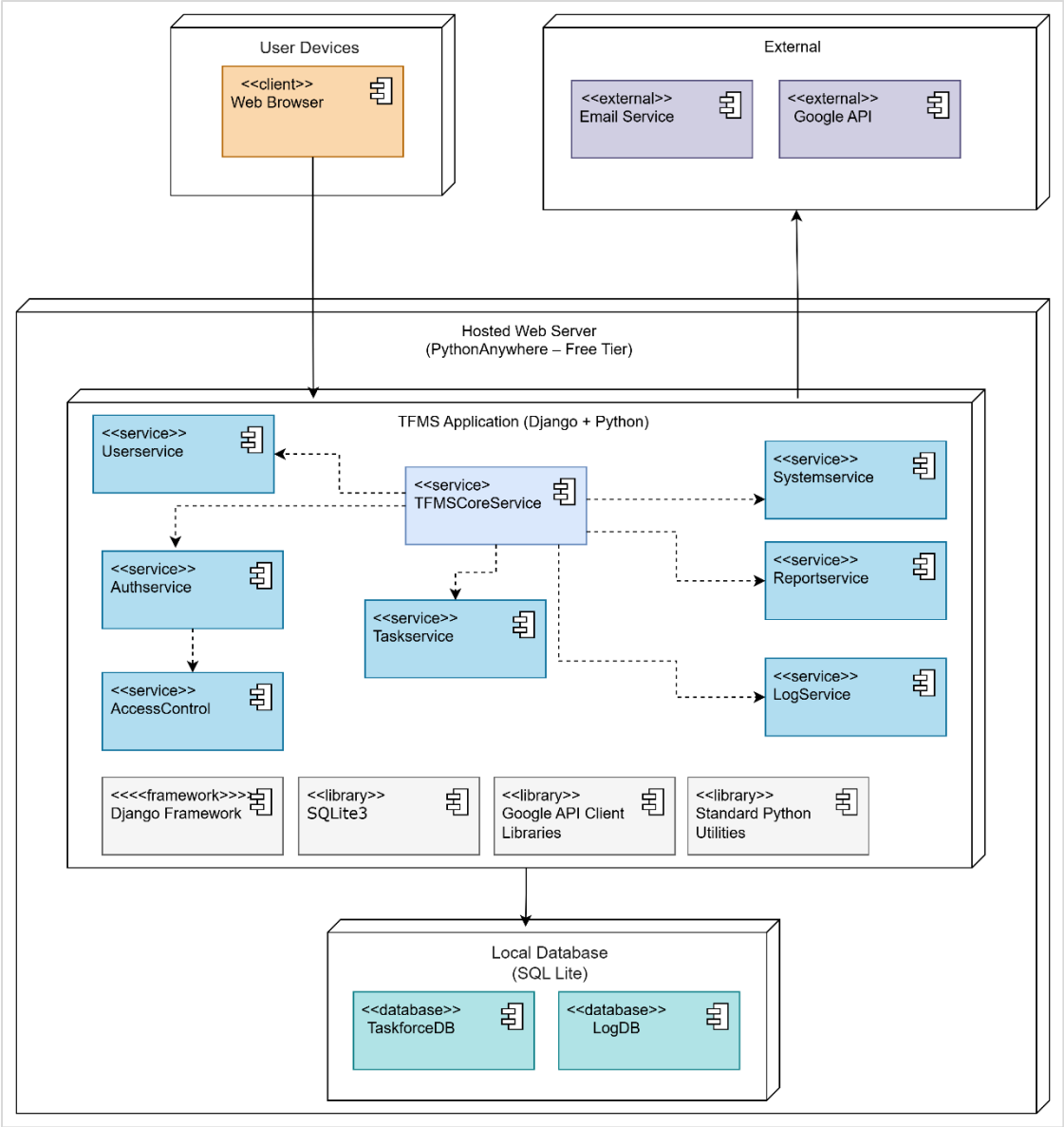


Figure 11: Deployment Diagram of TFMS

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3.5 Dynamic Organization

This section describes dynamic organization, communications, and interoperations of the TFMS components using sequence diagram. The following sequence diagrams are drawn based on TFMS use case diagrams and it is organized into basic flow, alternative flow and exception flow. The TFMS has 6 number sequence diagrams. These diagrams show the interaction between classes or objects, according to basic and alternative flows of TFMS use cases that mentioned in SRS document.

3.5.1 Sequence Diagram for All Users

Refer to Annex 1, Appendix 1: Sequence Diagram - User Authentication and Login

3.5.2 Sequence Diagram for System Administrator

Refer to Annex 1, Appendix 2: Sequence Diagram - Configure TFMS Parameters and Master Data Management

3.5.3 Sequence Diagram for HOD/Task Force Owner

Refer to Annex 1, Appendix 3: Sequence Diagram - Manage Task Force Membership and Workload Distribution

3.5.4 Sequence Diagram PSM/HR Office

Refer to Annex 1, Appendix 4: Sequence Diagram - Review and Approve Task Force Assignments

3.5.5 Sequence Diagram Lecturer

Refer to Annex 1, Appendix 5: Sequence Diagram - Access Personal Task Force Portfolio

3.5.6 Sequence Diagram Faculty Management

Refer to Annex 1, Appendix 6: Sequence Diagram - Generate Executive Reports and Analytics



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3.6 State Transition Diagram

Figure 12 illustrates state transition Diagram of the TFMS. The system begins in an idle state and transitions to authentication when a user attempts login. Valid credentials grant access to various functionalities such as system configuration, task assignment, portfolio viewing, and report generation, depending on user roles (Admin, HOD, Lecturer, Management). Invalid login attempts may lead to repeated authentication or account lockout. Post-approval of assignments, lecturers can view their portfolios, and management can generate reports. All user activities eventually transition to the logged-out state, returning the system to idle.

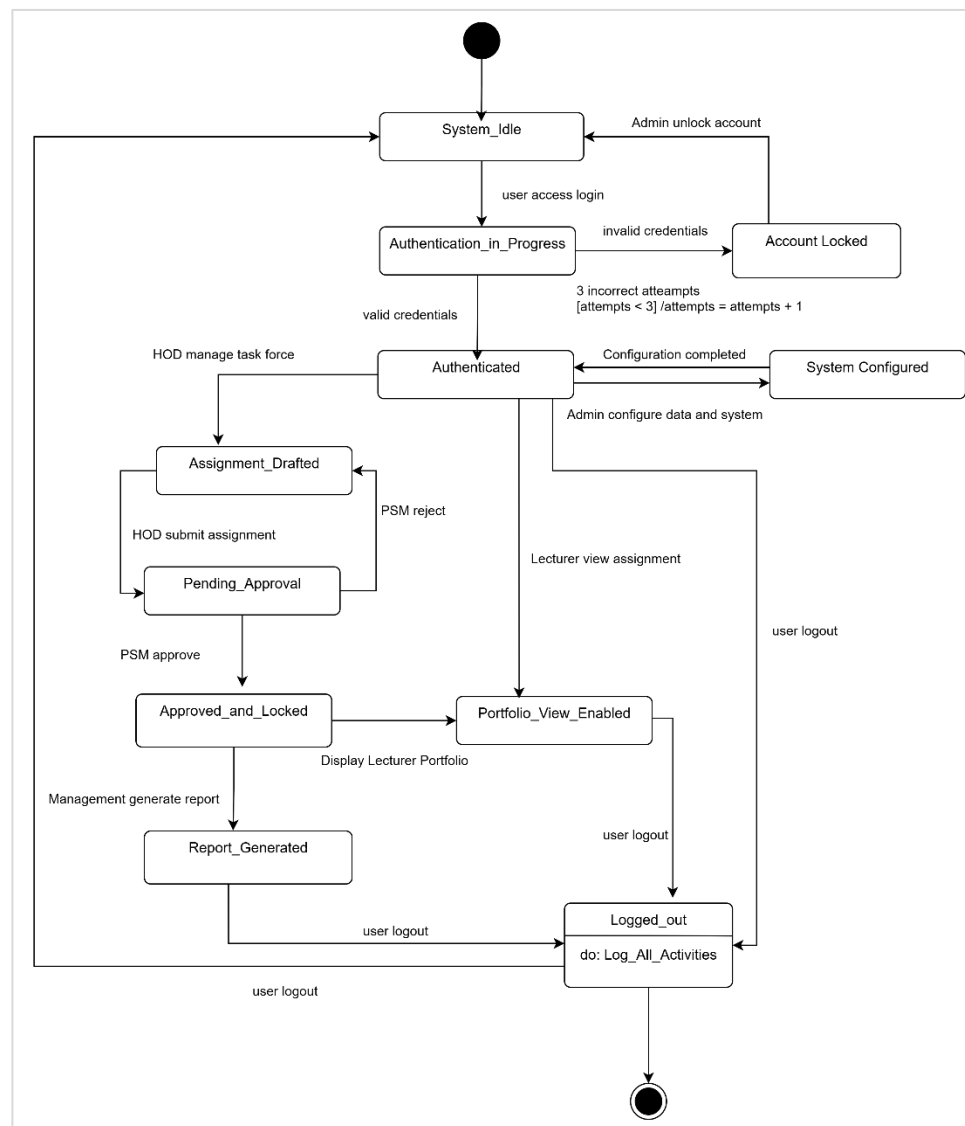


Figure 12: State Transition Diagram of TFMS



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3.7 Application Workflows

3.7.1 User Authentication and Login

Figure 13 illustrates the activity flow for the User Authentication and Login process, delineating the dynamic interactions between the User and the TFMS system across two distinct swim lanes.

- 1. Authentication Logic:** The process initiates when the user navigates to the login page and submits credentials. The system evaluates these inputs through a series of conditional branches, prioritizing security checks. It first validates the credential’s structure and correctness before assessing the account's standing.
- 2. Security Constraints (Lockout and Validation):** The diagram details the exception handling for invalid login attempts. The system implements a counter mechanism; if the number of consecutive failed attempts reaches a predefined threshold, the system transitions the account to a "Locked" state to prevent brute-force attacks. Otherwise, it logs the failure and permits a retry.
- 3. Account Management Flows:**
 - a) **First-Time Login:** Upon successful validation, the system checks a "First Login" flag. If active, the user is diverted to a mandatory password change workflow before access is granted.
 - b) **Password/Account Recovery:** A parallel branch handles the "Forgot Password" request, triggering an asynchronous recovery process involving token generation, email distribution, and secure password reset validation.
- 4. Session Initialization:** In the standard successful scenario, once all security conditions (valid credentials, unlocked account, established password) are met, the system creates a user session, logs the successful entry, and redirects the user to their role-specific dashboard.

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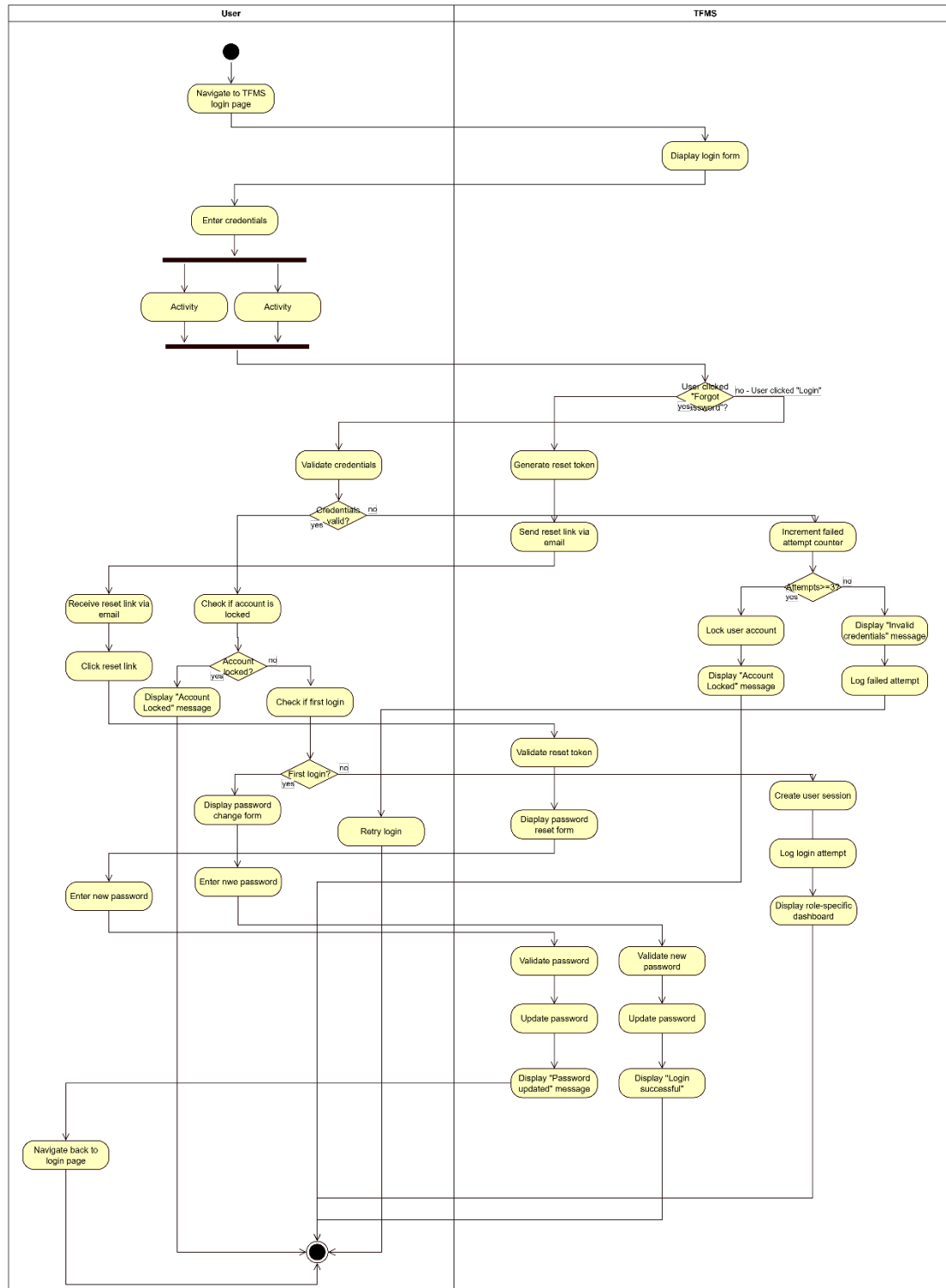


Figure 13: Activity Diagrams - User Authentication and Login



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3.7.2 Configure TFMS Parameters and Master Data Management

Figure 14 depicts the administrative workflows managed by the System Administrator, showcasing parallel paths for managing Staff, Task Forces, and System Configuration.

- 1. Administration Dashboard Entry:** The workflow begins with the administrator logging in and accessing the main control panel. From here, the system presents three distinct operational branches: Staff Management, Task Force Management, and Configuration.
- 2. Staff Management Branch:**
 - a) **CRUD Operations:** The diagram shows the flow for adding new staff and updating existing records.
 - b) **Access Control:** Specific paths are defined for unlocking accounts (resetting failed attempts) and deactivating staff members (requiring justification), directly mapping to the security requirements.
- 3. Task Force Management Branch:** This path handles the structural organization of the institution. The administrator can create new task forces, defining their scope and parameters, which the system then persists in the database.
- 4. System Configuration Branch:** The third branch illustrates the maintenance of global system rules. The administrator can modify parameters (like workload thresholds), which the system validates and consistently applies across all modules.
- 5. Audit Logging:** A crucial parallel process, indicated by the continuous flow to "Log Action," ensures that every administrative change, whether creating a user, modifying a task force, or changing a rule is recorded in the global audit trail before the use case concludes.



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SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	SDD-TFMS-190126-V1-	A4	1.0	35/101

3.7.3 Manage Task Force Membership and Workload Distribution

Figure 15 outlines the workflow for the Head of Department (HOD) in managing task force memberships, highlighting the interaction between decision-making and system validation.

- 1. Initialization:** The process starts with the HOD logging in and accessing the Task Force Management module. The system retrieves and displays the department's relevant task forces along with their current membership rosters.
- 2. Membership Modification:** The HOD proceeds to select a specific task force and modifies its composition by adding or removing members. The system immediately validates the eligibility of the selected staff (ensuring they are "Active").
- 3. Workload Validation Loop:**
 - a) **Impact Calculation:** Upon modification, the system calculates the projected workload impact.
 - b) **Overload Detection:** A critical decision node checks for "Workload Overload."
 - c) **Resolution Paths:**
 - a. **No Overload:** The process proceeds directly to submission.
 - b. **Overload Detected:** The system triggers a warning. The HOD must then choose to either Adjust Assignments (looping back to recalculate) or Override.
- 4. Submission:** Once the membership is finalized, the HOD submits the configuration. The system then routes the request to the PSM and logs the submission, completing the activity.



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	SDD-TFMS-190126-V1-	A4	1.0	37/101

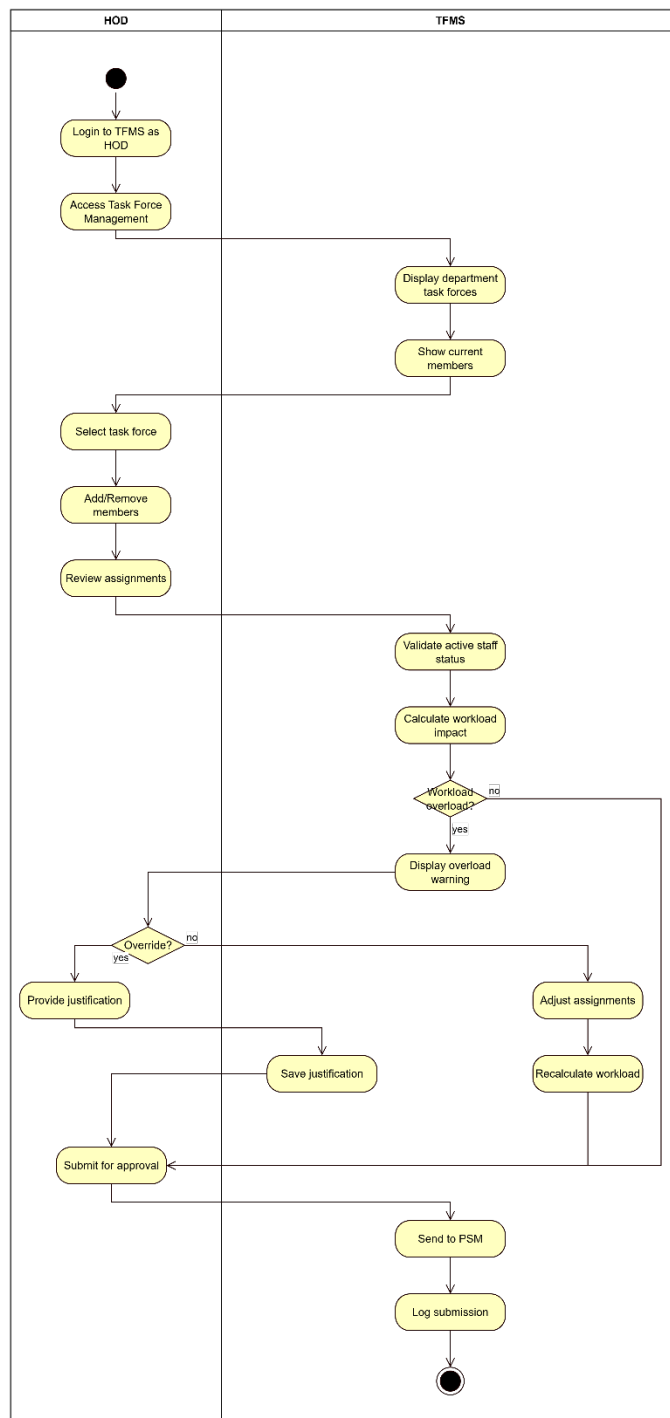


Figure 15: Activity Diagram - Manage Task Force Membership and Workload Distribution



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	SDD-TFMS-190126-V1-	A4	1.0	38/101

3.7.4 Review and Approve Task Force Assignments

Figure 16 illustrates the vetting workflow undertaken by the PSM to ensure task force compliance, involving a cycle of review, adjustment, and final decision.

- 1. Review Initialization:** The activity begins with the PSM navigating to the Approval Panel. The system automatically queries for pending submissions. If the queue is empty, the process terminates early; otherwise, it presents a list of departmental submissions for detailed inspection.
- 2. Data-Driven Review:** For a selected submission, the system not only shows the staffing plan but also pulls "Workload Metrics" and "Inventory" data. This aligns with the requirement for informed decision-making based on resource availability.
- 3. Adjustment Cycle:** The diagram highlights a unique capability where the PSM can intervene directly. If the PSM determines "Need Adjustments" (e.g., to balance equity), they can modify assignments. Crucially, any modification triggers a mandatory "Recalculate Workloads" system action to ensure the new configuration remains valid.
- 4. Approval Decision:**
 - a) **Rejection:** If the submission is fundamentally flawed, the PSM enters a "Rejection Reason," and the system records this negative outcome.
 - b) **Approval:** If accepted (either originally or after adjustment), the system "Locks Assignments" to prevent further changes, formally records the approval, and triggers notifications to the HOD, marking the successful conclusion of the assignment phase.



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	SDD-TFMS-190126-V1-	A4	1.0	39/101

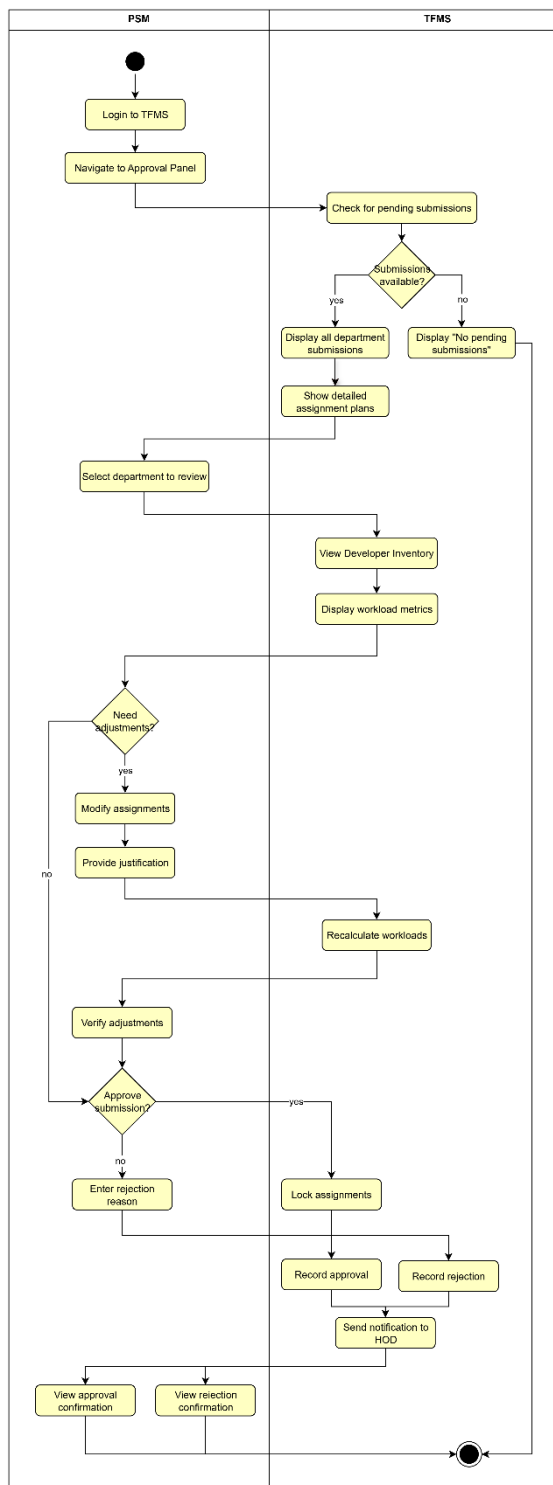


Figure 16: Activity Diagram - Review and Approve Task Force Assignments



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	SDD-TFMS-190126-V1-	A4	1.0	40/101

3.7.5 Access Personal Task Force Portfolio

Figure 17 details the activities associated with the Lecturer's personal portfolio, encompassing data retrieval, reporting, and feedback mechanisms.

- 1. Portfolio Retrieval:** The workflow begins with the Lecturer accessing the "My Portfolio" section. The system verifies the existence of current assignments. If none are found, it defaults to displaying a "No assignments" message while still allowing access to historical records.
- 2. Dashboard Visualization:** For active lecturers, the system populates the dashboard with "Task Forces," "Roles," and "Weightage" data, providing an immediate snapshot of their current academic service load.
- 3. Reporting and Feedback:**
 - a) **Data Export:** The diagram illustrates the decision path for downloading summaries. If initiated, the system triggers a PDF generation process (`generateReport()`), checks for success, and provides a download link.
- 4. Submission of Remarks:** A distinct feature shown is the ability for lecturers to "Submit Remarks," which the system captures, notifies the relevant HOD/PSM about, and permanently records, facilitating bidirectional communication regarding assignments.



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	SDD-TFMS-190126-V1-	A4	1.0	41/101

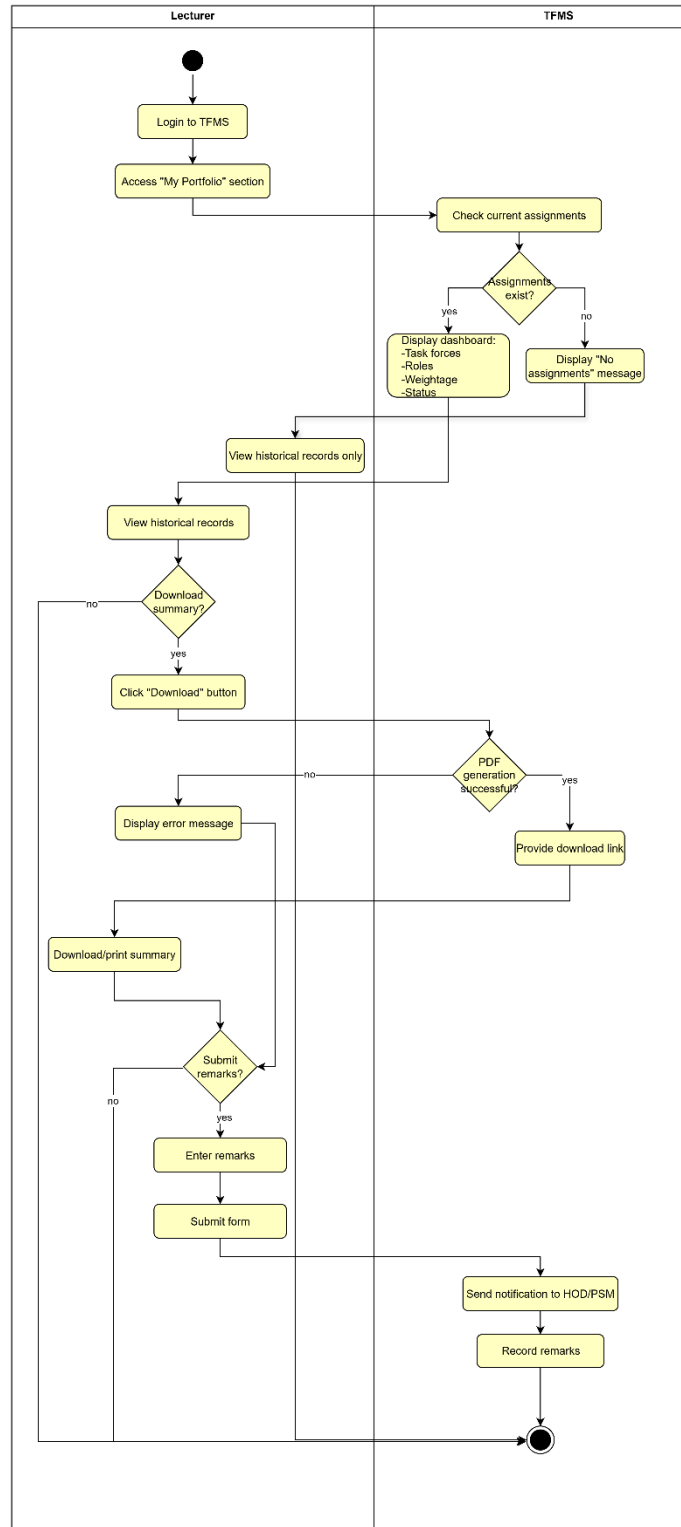


Figure 17: Activity Diagram - Access Personal Task Force Portfolio



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	SDD-TFMS-190126-V1-	A4	1.0	42/101

3.7.6 Generate Executive Reports and Analytics

Figure 18 details the strategic oversight workflow for Faculty Management (e.g., Dean), emphasizing data visualization, drill-down analysis, and formal report generation.

- 1. **Executive Dashboard Initialization:** Upon accessing the "Executive Reports" module, the system loads a high-level dashboard. Crucially, this view aggregates data to present "Distribution Charts," "Fairness Metrics," and "Department Comparisons," offering an immediate strategic overview.
- 2. **Multi-Level Drill-Down:** The diagram illustrates a hierarchical analysis capability. Management can choose to "Need detailed analysis," which allows them to navigate from the faculty level down to the "Department Level," and further down to the "Individual Level" (specific lecturer) to scrutinize granular assignment details.
- 3. **Flexible Report Generation:**
 - a) **Format Selection:** The decision node "Generate report?" leads to a configuration phase where the user selects the output format (PDF/Excel) and the scope of the data.
 - b) **System Processing:** The system generates the report. If successful, it not only provides the download but also executes a governance protocol: notifications are sent to relevant HODs/PSMs to alert them of the executive review, and any remarks acting as feedback are permanently recorded.

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SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	SDD-TFMS-190126-V1-	A4	1.0	43/101

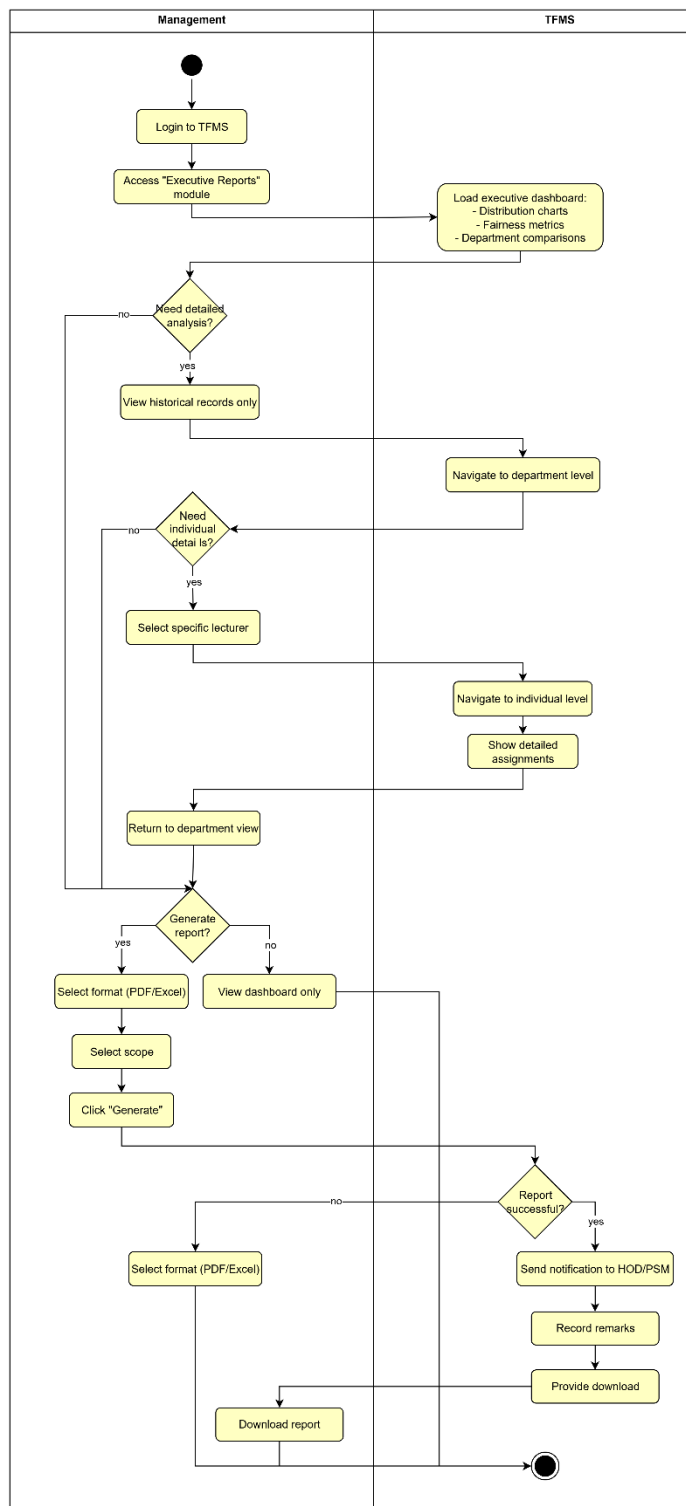


Figure 18: Activity Diagram - Generate Executive Reports and Analytics



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	SDD-TFMS-190126-V1-	A4	1.0	44/101

3.8 CSCs and CSUs Design Description

This section describes the information about each CSCs and CSUs of TFMS, including their identification, type, attributes and operations.

3.8.1 CSC LoginView (TFMS-01-100-PY-CSC-1)

This CSC consists of two CSUs which are UI and User.

3.8.1.1 CSU UI (TFMS-01-100-PY-CSU-1.1)

The table below shows the design description of CSU UI.

Table 4: CSU UI Design Description

Type	Boundary	
Description	Represents the user interface component for the LoginView.	
Attributes		
	Name	Type
	field	Type
Operation		
	Name	Return Type
	displayDashboard()	Void
	submitForm()	Boolean
	showNotification()	Void

3.8.1.2 CSU User (TFMS-01-100-PY-CSU-1.2)

The table below shows the design description of CSU User.

Table 5: CSU User Design Description

Type	Entity	
Description	Represents a user in the system with authentication and profile management capabilities.	
Attributes		
	Name	Type
	staffID	String
	staffName	String
	userName	String
	password	String
	department	String
	grade	String



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	SDD-TFMS-190126-V1-	A4	1.0	45/101

	category isActive	String Boolean
Operation		
	Name viewProfile() login() logout() resetPassword() lockAccount()	Return Type UserProfile Boolean Void Boolean Void

3.8.2 CSC LecturerDashboard (TFMS-02-100-PY-CSC-2)

This CSC of two CSUs which are UI and User.

3.8.2.1 CSU UI (TFMS-02-100-PY-CSU-2.1)

The table below shows the design description of CSU UI.

Table 6: CSU UI Design Description

Type	Boundary	
Description	Represents the user interface component for the Lecturer Dashboard.	
Attributes		
	Name	Type
	dashboardLayout	String
Operation		
	Name	Return Type
	displayDashboard()	Void
	showWorkloadSummary()	Void
	showTaskForcePortfolio()	Void

3.8.2.2 CSU User (TFMS-02-100-PY-CSU-2.2)

The table below shows the design description of CSU User.

Table 7: CSU User Design Description

Type	Entity
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DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	SDD-TFMS-190126-V1-	A4	1.0	46/101

Description	Represents a user in the system with authentication and profile management capabilities.																			
Attributes	<table><tr><th>Name</th><th>Type</th></tr><tr><td>staffID</td><td>String</td></tr><tr><td>staffName</td><td>String</td></tr><tr><td>userName</td><td>String</td></tr><tr><td>Password</td><td>String</td></tr><tr><td>department</td><td>String</td></tr><tr><td>grade</td><td>String</td></tr><tr><td>category</td><td>String</td></tr><tr><td>isActive</td><td>Boolean</td></tr></table>		Name	Type	staffID	String	staffName	String	userName	String	Password	String	department	String	grade	String	category	String	isActive	Boolean
Name	Type																			
staffID	String																			
staffName	String																			
userName	String																			
Password	String																			
department	String																			
grade	String																			
category	String																			
isActive	Boolean																			
Operation	<table><tr><th>Name</th><th>Return Type</th></tr><tr><td>viewProfile()</td><td>UserProfile</td></tr><tr><td>login()</td><td>Boolean</td></tr><tr><td>logout()</td><td>Void</td></tr><tr><td>resetPassword()</td><td>Boolean</td></tr><tr><td>lockAccount()</td><td>Void</td></tr></table>		Name	Return Type	viewProfile()	UserProfile	login()	Boolean	logout()	Void	resetPassword()	Boolean	lockAccount()	Void						
Name	Return Type																			
viewProfile()	UserProfile																			
login()	Boolean																			
logout()	Void																			
resetPassword()	Boolean																			
lockAccount()	Void																			

3.8.3 CSC HODDashboard (TFMS-03-100-PY-CSC-3)

This CSC consists of two CSUs which are UI and User.

3.8.3.1 CSU UI (TFMS-03-100-PY-CSU-3.1)

The table below shows the design description of CSU UI.

Table 8: CSU UI Design Description

Type	Boundary	
Description	Represents the user interface component for the HOD Dashboard.	
Attributes		
	Name	Type
	departmentView	String
Operation		
	Name	Return Type
	displayDepartmentOverview() showWorkloadDistribution()	Void Void



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	SDD-TFMS-190126-V1-	A4	1.0	47/101

	manageTaskForceProposals()	Void
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3.8.3.2 CSU User (TFMS-03-100-PY-CSU-3.2)

The table below shows the design description of CSU User.

Table 9: CSU User Design Description

Type	Entity																			
Description	Represents a user in the system with authentication and profile management capabilities.																			
Attributes	<table><tr><th>Name</th><th>Type</th></tr><tr><td>staffID</td><td>String</td></tr><tr><td>staffName</td><td>String</td></tr><tr><td>userName</td><td>String</td></tr><tr><td>Password</td><td>String</td></tr><tr><td>department</td><td>String</td></tr><tr><td>grade</td><td>String</td></tr><tr><td>category</td><td>String</td></tr><tr><td>isActive</td><td>Boolean</td></tr></table>		Name	Type	staffID	String	staffName	String	userName	String	Password	String	department	String	grade	String	category	String	isActive	Boolean
Name	Type																			
staffID	String																			
staffName	String																			
userName	String																			
Password	String																			
department	String																			
grade	String																			
category	String																			
isActive	Boolean																			
Operation	<table><tr><th>Name</th><th>Return Type</th></tr><tr><td>viewProfile()</td><td>UserProfile</td></tr><tr><td>login()</td><td>Boolean</td></tr><tr><td>logout()</td><td>Void</td></tr><tr><td>resetPassword()</td><td>Boolean</td></tr><tr><td>lockAccount()</td><td>Void</td></tr></table>		Name	Return Type	viewProfile()	UserProfile	login()	Boolean	logout()	Void	resetPassword()	Boolean	lockAccount()	Void						
Name	Return Type																			
viewProfile()	UserProfile																			
login()	Boolean																			
logout()	Void																			
resetPassword()	Boolean																			
lockAccount()	Void																			

3.8.4 CSC PSMDashboard (TFMS-04-100-PY-CSC-4)

This CSC consists of three CSUs which are UI, User and ApprovalRequest.

3.8.4.1 CSU UI (TFMS-04-100-PY-CSU-4.1)

The table below shows the design description of CSU UI.

Table 10: CSU UI Design Description

Type	Boundary
Description	Represents the user interface component for the PSM/HR Dashboard.



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SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	SDD-TFMS-190126-V1-	A4	1.0	48/101

Attributes		
	Name	Type
	approvalQueue	List
Operation		
	Name	Return Type
	displayApprovalDashboard()	Void
	showFacultyBalancing()	Void
	processApprovalWorkflow()	Boolean

3.8.4.2 CSU User (TFMS-04-100-PY-CSU-4.2)

The table below shows the design description of CSU User.

Table 11: CSU User Design Description

Type	Entity																			
Description	Represents a user in the system with authentication and profile management capabilities.																			
Attributes	<table><tr><th>Name</th><th>Type</th></tr><tr><td>staffID</td><td>String</td></tr><tr><td>staffName</td><td>String</td></tr><tr><td>userName</td><td>String</td></tr><tr><td>password</td><td>String</td></tr><tr><td>department</td><td>String</td></tr><tr><td>grade</td><td>String</td></tr><tr><td>category</td><td>String</td></tr><tr><td>isActive</td><td>Boolean</td></tr></table>		Name	Type	staffID	String	staffName	String	userName	String	password	String	department	String	grade	String	category	String	isActive	Boolean
Name	Type																			
staffID	String																			
staffName	String																			
userName	String																			
password	String																			
department	String																			
grade	String																			
category	String																			
isActive	Boolean																			
Operation	<table><tr><th>Name</th><th>Return Type</th></tr><tr><td>viewProfile()</td><td>UserProfile</td></tr><tr><td>login()</td><td>Boolean</td></tr><tr><td>logout()</td><td>Void</td></tr><tr><td>resetPassword()</td><td>Boolean</td></tr><tr><td>lockAccount()</td><td>Void</td></tr></table>		Name	Return Type	viewProfile()	UserProfile	login()	Boolean	logout()	Void	resetPassword()	Boolean	lockAccount()	Void						
Name	Return Type																			
viewProfile()	UserProfile																			
login()	Boolean																			
logout()	Void																			
resetPassword()	Boolean																			
lockAccount()	Void																			

3.8.4.3 CSU ApprovalRequest (TFMS-04-100-PY-CSU-4.3)

The table below shows the design description of CSU ApprovalRequest.



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	SDD-TFMS-190126-V1-	A4	1.0	49/101

Table 12: CSU ApprovalRequest Design Description

Type	Entity															
Description	Represents an approval request for task force assignments.															
Attributes	<table><tr><th>Name</th><th>Type</th></tr><tr><td>approvalId</td><td>String</td></tr><tr><td>submissionDate</td><td>Datetime</td></tr><tr><td>status</td><td>String</td></tr><tr><td>remarks</td><td>String</td></tr><tr><td>decisionDate</td><td>Datetime</td></tr><tr><td>approverId</td><td>String</td></tr></table>		Name	Type	approvalId	String	submissionDate	Datetime	status	String	remarks	String	decisionDate	Datetime	approverId	String
Name	Type															
approvalId	String															
submissionDate	Datetime															
status	String															
remarks	String															
decisionDate	Datetime															
approverId	String															
Operation	<table><tr><th>Name</th><th>Return Type</th></tr><tr><td>approve()</td><td>Boolean</td></tr><tr><td>reject()</td><td>Boolean</td></tr><tr><td>escalate()</td><td>Boolean</td></tr></table>		Name	Return Type	approve()	Boolean	reject()	Boolean	escalate()	Boolean						
Name	Return Type															
approve()	Boolean															
reject()	Boolean															
escalate()	Boolean															

3.8.5 CSC ManagementDashboard (TFMS-05-100-PY-CSC-5)

This CSC consists of two CSUs which are UI and User.

3.8.5.1 CSU UI (TFMS-05-100-PY-CSU-5.1)

The table below shows the design description of CSU UI.

Table 13: CSU UI Design Description

Type	Boundary	
Description	Represents the user interface component for the Faculty Management Dashboard.	
Attributes		
	Name	Type
	strategicMetrics	Dictionary
Operation		
	Name	Return Type
	displayExecutiveDashboard()	Void
	showStrategicSummaries()	Void
	generateAnalyticsView()	Void

**DOCUMENT IDENTIFICATION**

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	SDD-TFMS-190126-V1-	A4	1.0	50/101

3.8.5.2 CSU User (TFMS-05-100-PY-CSU-5.2)

The table below shows the design description of CSU User.

Table 14: CSU User Design Description

Type	Entity																			
Description	Represents a user in the system with authentication and profile management capabilities.																			
Attributes	<table><tr><th>Name</th><th>Type</th></tr><tr><td>staffID</td><td>String</td></tr><tr><td>staffName</td><td>String</td></tr><tr><td>userName</td><td>String</td></tr><tr><td>password</td><td>String</td></tr><tr><td>department</td><td>String</td></tr><tr><td>grade</td><td>String</td></tr><tr><td>category</td><td>String</td></tr><tr><td>isActive</td><td>Boolean</td></tr></table>		Name	Type	staffID	String	staffName	String	userName	String	password	String	department	String	grade	String	category	String	isActive	Boolean
Name	Type																			
staffID	String																			
staffName	String																			
userName	String																			
password	String																			
department	String																			
grade	String																			
category	String																			
isActive	Boolean																			
Operation	<table><tr><th>Name</th><th>Return Type</th></tr><tr><td>viewProfile()</td><td>UserProfile</td></tr><tr><td>login()</td><td>Boolean</td></tr><tr><td>logout()</td><td>Void</td></tr><tr><td>resetPassword()</td><td>Boolean</td></tr><tr><td>lockAccount()</td><td>Void</td></tr></table>		Name	Return Type	viewProfile()	UserProfile	login()	Boolean	logout()	Void	resetPassword()	Boolean	lockAccount()	Void						
Name	Return Type																			
viewProfile()	UserProfile																			
login()	Boolean																			
logout()	Void																			
resetPassword()	Boolean																			
lockAccount()	Void																			

3.8.6 CSC TFMSCoreService (TFMS-06-100-PY-CSC-6)

This CSC consists of one CSU which is TFMSController.

3.8.6.1 CSU TFMSController (TFMS-06-100-PY-CSU-6.1)

The table below shows the design description of CSU TFMSController.

Table 15: CSU TFMSController Design Description

Type	Entity			
Description	Central controller coordinating system workflows and business logic.			
Attributes	<table><tr><th>Name</th><th>Type</th></tr></table>		Name	Type
Name	Type			

**DOCUMENT IDENTIFICATION**

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	SDD-TFMS-190126-V1-	A4	1.0	51/101

	serviceRegistry		Dictionary
Operation			
	Name		Return Type
	authenticateUser()		Boolean
	manageTaskForce()		Boolean
	manageMembership()		Boolean
	calculateWorkload()		Float
	processApproval()		Boolean
	generateReport()		Report
coordinateServices()		Void	

3.8.7 CSC UserService (TFMS-07-100-PY-CSC-7)

This CSC consists of three CSUs which are User, Role, and UserRole.

3.8.7.1 CSU User (TFMS-07-100-PY-CSU-7.1)

The table below shows the design description of CSU User.

Table 16: CSU User Design Description

Type	Entity																			
Description	Represents a user in the system with authentication and profile management capabilities.																			
Attributes	<table><tr><th>Name</th><th>Type</th></tr><tr><td>staffID</td><td>String</td></tr><tr><td>staffName</td><td>String</td></tr><tr><td>userName</td><td>String</td></tr><tr><td>password</td><td>String</td></tr><tr><td>department</td><td>String</td></tr><tr><td>grade</td><td>String</td></tr><tr><td>category</td><td>String</td></tr><tr><td>isActive</td><td>Boolean</td></tr></table>		Name	Type	staffID	String	staffName	String	userName	String	password	String	department	String	grade	String	category	String	isActive	Boolean
Name	Type																			
staffID	String																			
staffName	String																			
userName	String																			
password	String																			
department	String																			
grade	String																			
category	String																			
isActive	Boolean																			
Operation	<table><tr><th>Name</th><th>Return Type</th></tr><tr><td>viewProfile()</td><td>UserProfile</td></tr><tr><td>login()</td><td>Boolean</td></tr><tr><td>logout()</td><td>Void</td></tr><tr><td>resetPassword()</td><td>Boolean</td></tr><tr><td>lockAccount()</td><td>Void</td></tr></table>		Name	Return Type	viewProfile()	UserProfile	login()	Boolean	logout()	Void	resetPassword()	Boolean	lockAccount()	Void						
Name	Return Type																			
viewProfile()	UserProfile																			
login()	Boolean																			
logout()	Void																			
resetPassword()	Boolean																			
lockAccount()	Void																			



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Task Force Management System (TFMS)	SDD-TFMS-190126-V1-	A4	1.0	52/101

3.8.7.2 CSU Role (TFMS-07-100-PY-CSU-7.2)

The table below shows the design description of CSU Role.

Table 17: CSU Role Design Description

Type	Entity	
Description	Represents a system role with access permissions and privileges.	
Attributes		
Operation		

3.8.7.3 CSU UserRole (TFMS-07-100-PY-CSU-7.3)

The table below shows the design description of CSU UserRole

Table 18: CSU UserRole Design Description

Type	Entity	
Description	Represents the association between a User and a Role.	
Attributes		
	Name	Type
	isPrimary	Boolean
	assignmentDate	DateTime
	assignedBy	String
Operation		
	Name	Return Type
	assignRole()	Boolean
	revokeRole()	Boolean
	setAsPrimary()	Void

**DOCUMENT IDENTIFICATION**

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	SDD-TFMS-190126-V1-	A4	1.0	53/101

3.8.8 CSC TaskService (TFMS-08-100-PY-CSC-8)

This CSC consists of four CSUs which are TaskForce, TaskForceMembership, Department, and WeightageRule.

3.8.8.1 CSU TaskForce (TFMS-08-100-PY-CSU-8.1)

The table below shows the design description of CSU TaskForce.

Table 19: CSU TaskForce Design Description

Type	Entity																	
Description	Represents a task force entity within the system.																	
Attributes	<table><tr><th>Name</th><th>Type</th></tr><tr><td>taskForceId</td><td>String</td></tr><tr><td>name</td><td>String</td></tr><tr><td>description</td><td>String</td></tr><tr><td>defaultWeightage</td><td>Float</td></tr><tr><td>startDate</td><td>DateTime</td></tr><tr><td>endDate</td><td>DateTime</td></tr><tr><td>status</td><td>String</td></tr></table>		Name	Type	taskForceId	String	name	String	description	String	defaultWeightage	Float	startDate	DateTime	endDate	DateTime	status	String
	Name	Type																
	taskForceId	String																
	name	String																
	description	String																
	defaultWeightage	Float																
	startDate	DateTime																
	endDate	DateTime																
status	String																	
Operation	<table><tr><th>Name</th><th>Return Type</th></tr><tr><td>activate()</td><td>Boolean</td></tr><tr><td>deactivate()</td><td>Boolean</td></tr><tr><td>addMember()</td><td>Boolean</td></tr></table>		Name	Return Type	activate()	Boolean	deactivate()	Boolean	addMember()	Boolean								
	Name	Return Type																
	activate()	Boolean																
	deactivate()	Boolean																
addMember()	Boolean																	

3.8.8.2 CSU TaskForceMembership (TFMS-08-100-PY-CSU-8.2)

The table below shows the design description of CSU TaskForceMembership.

Table 20: CSU TaskForceMembership Design Description

Type	Entity	
Description	Represents a lecturer's membership in a task force.	
Attributes		
	Name	Type
	membershipId	String



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SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	SDD-TFMS-190126-V1-	A4	1.0	54/101

	roleInTaskForce	String
	weightage	Float
	startDate	DateTime
	endDate	DateTime
	justification	String
	status	String
Operation		
	Name	Return Type
	calculateContribution() updateStatus()	Float Boolean

3.8.8.3 CSU Department (TFMS-08-100-PY-CSU-8.3)

The table below shows the design description of CSU Department.

Table 21: CSU Department Design Description

Type	Entity	
Description	Represents a department within the faculty.	
Attributes	Name	Type
	code	String
	name	String
	faculty	String
	headId	String
Operation	Name	Return Type
	getStaffList()	List
	calculateDepartmentWorkload()	Float
	updateDepartmentInfo()	Boolean

3.8.8.4 CSU WeightageRule (TFMS-08-100-PY-CSU-8.4)

The table below shows the design description of CSU WeightageRule.

Table 22: CSU WeightageRule Design Description

	DOCUMENT IDENTIFICATION				
	SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
	Task Force Management System (TFMS)	SDD-TFMS-190126-V1-	A4	1.0	55/101

Type	Entity	
Description	Defines weightage rules for task force categories and roles.	
Attributes		
	Name	Type
	category	String
	min_Weightage	Float
	max_Weightage	Float
	applicableRoles	List
Operation		
	Name	Return Type
	validateWeightage()	Boolean
	getWeightageRange()	Dictionary
	updateRule()	Boolean

3.8.9 CSC AuthService (TFMS-09-100-PY-CSC-9)

This CSC consists of two CSUs which are User and Role.

3.8.9.1 CSU User (TFMS-09-100-PY-CSU-9.1)

The table below shows the design description of CSU User.

Table 23: CSU User Design Description

Type	Entity	
Description	Represents a user in the system with authentication and profile management capabilities.	
Attributes		
	Name	Type
	staffID	String
	staffName	String
	userName	String
	password	String
	department	String
	grade	String
	category	String
isActive	Boolean	
Operation		
	Name	Return Type
	viewProfile()	UserProfile



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	SDD-TFMS-190126-V1-	A4	1.0	56/101

	login() logout() resetPassword() lockAccount()	Boolean Void Boolean Void	
--	---	------------------------------------	--

3.8.9.2 CSU Role (TFMS-09-100-PY-CSU-9.2)

The table below shows the design description of CSU Role.

Table 24: CSU Role Design Description

Type	Entity		
Description	Represents a system role with access permissions and privileges.		
Attributes	Name		Type
	name		String
	code		String
	permissions		List
	description		String
Operation	Name		Return Type
	addPermission()		Boolean
	removePermission()		Boolean
	validateAccess()		Boolean

3.8.10 CSC ReportService (TFMS-10-100-PY-CSC-10)

This CSC consists of one CSU which is Report.

3.8.10.1 CSU Report (TFMS-10-100-PY-CSU-10.1)

The table below shows the design description of CSU Report.

Table 25: CSU Report Design Description

Type	Entity		
Description	Represents a report entity with generation and export capabilities.		
Attributes	Name	Type	
	reportType	String	



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	SDD-TFMS-190126-V1-	A4	1.0	57/101

	reportDate	DateTime
	reportId	String
Operation		
	Name	Return Type
	generateReport()	Report
	exportReportPDF()	Boolean
	exportReportExcel()	Boolean

3.8.11 CSC SystemService (TFMS-11-100-PY-CSC-11)

This CSC consists of one CSU which is SystemConfig.

3.8.11.1 CSU SystemConfig (TFMS-11-100-PY-CSU-11.1)

The table below shows the design description of CSU SystemConfig.

Table 26: CSU SystemConfig Design Description

Type	Entity	
Description	Represents system configuration parameters and settings.	
Attributes	Name	Type
	config_key	String
	config_value	String
	description	String
	lastModified	DateTime
	modifiedBy	String
Operation	Name	Return Type
	getConfig()	String
	setConfig()	Boolean
	listAllConfigs()	Dictionary

3.8.12 CSC LogService (TFMS-12-100-PY-CSC-12)

This CSC consists of one CSU which is AuditLog.



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SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
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3.8.12.1 CSU AuditLog (TFMS-12-100-PY-CSU-12.1)

The table below shows the design description of CSU AuditLog.

Table 27: CSU AuditLog Design Description

Type	Entity																			
Description	Represents audit log entries for system activities and compliance tracking.																			
Attributes	<table><tr><th>Name</th><th>Type</th></tr><tr><td>logId</td><td>String</td></tr><tr><td>actionType</td><td>String</td></tr><tr><td>entityName</td><td>String</td></tr><tr><td>entityId</td><td>String</td></tr><tr><td>timeStamp</td><td>String</td></tr><tr><td>userId</td><td>String</td></tr><tr><td>justification</td><td>String</td></tr><tr><td>ipAddress</td><td>Boolean</td></tr></table>		Name	Type	logId	String	actionType	String	entityName	String	entityId	String	timeStamp	String	userId	String	justification	String	ipAddress	Boolean
	Name	Type																		
	logId	String																		
	actionType	String																		
	entityName	String																		
	entityId	String																		
	timeStamp	String																		
	userId	String																		
	justification	String																		
ipAddress	Boolean																			
Operation	<table><tr><th>Name</th><th>Return Type</th></tr><tr><td>recordEvent()</td><td>Boolean</td></tr><tr><td>searchLogs()</td><td>List</td></tr><tr><td>exportLogs()</td><td>Boolean</td></tr></table>		Name	Return Type	recordEvent()	Boolean	searchLogs()	List	exportLogs()	Boolean										
	Name	Return Type																		
	recordEvent()	Boolean																		
	searchLogs()	List																		
exportLogs()	Boolean																			

3.8.13 CSC TaskforceRepository (TFMS-13-100-PY-CSC-13)

This CSC consists of three CSUs which are User, Role, and UserRole.

3.8.13.1 CSU User (TFMS-13-100-PY-CSU-13.1)

The table below shows the design description of CSU User.

Table 28: CSU User Design Description

Type	Entity	
Description	Represents a user in the system with authentication and profile management capabilities.	
Attributes		
	Name	Type
	staffID	String
	staffName	String
	userName	String

**DOCUMENT IDENTIFICATION**

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
Task Force Management System (TFMS)	SDD-TFMS-190126-V1-	A4	1.0	59/101

	password	String
	department	String
	grade	String
	category	String
	isActive	Boolean
Operation		
	Name	Return Type
	viewProfile()	UserProfile
	login()	Boolean
	logout()	Void
	resetPassword()	Boolean
	lockAccount()	Void

3.8.13.2 CSU Role (TFMS-13-100-PY-CSU-13.2)

The table below shows the design description of CSU Role.

Table 29: CSU Role Design Description

Type	Entity	
Description	Represents a system role with access permissions and privileges.	
Attributes	Name	Type
	name	String
	code	String
	permissions	List
	description	String
Operation	Name	Return Type
	addPermission()	Boolean
	removePermission()	Boolean
	validateAccess()	Boolean

3.8.13.3 CSU UserRole (TFMS-13-100-PY-CSU-13.3)

The table below shows the design description of CSU UserRole.

Table 30: CSU UserRole Design Description

	DOCUMENT IDENTIFICATION				
	SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
	Task Force Management System (TFMS)	SDD-TFMS-190126-V1-	A4	1.0	60/101

Type	Entity	
Description	Represents the association between a User and a Role.	
Attributes		
	Name	Type
	isPrimary	Boolean
	assignmentDate	DateTime
	assignedBy	String
Operation		
	Name	Return Type
	assignRole()	Boolean
	revokeRole()	Boolean
	setAsPrimary()	Void

3.8.14 CSC LogStorage (TFMS-14-100-PY-CSC-14)

This CSC consists of one CSU which is AuditLog.

3.8.14.1 CSU AuditLog (TFMS-14-100-PY-CSU-14.1)

The table below shows the design description of CSU AuditLog.

Table 31: CSU AuditLog Design Description

Type	Entity																			
Description	Represents audit log entries for system activities and compliance tracking.																			
Attributes	<table><tr><th>Name</th><th>Type</th></tr><tr><td>logId</td><td>String</td></tr><tr><td>actionType</td><td>String</td></tr><tr><td>entityName</td><td>String</td></tr><tr><td>entityId</td><td>String</td></tr><tr><td>timeStamp</td><td>DateTime</td></tr><tr><td>userId</td><td>String</td></tr><tr><td>justification</td><td>String</td></tr><tr><td>ipAddress</td><td>String</td></tr></table>		Name	Type	logId	String	actionType	String	entityName	String	entityId	String	timeStamp	DateTime	userId	String	justification	String	ipAddress	String
	Name	Type																		
	logId	String																		
	actionType	String																		
	entityName	String																		
	entityId	String																		
	timeStamp	DateTime																		
	userId	String																		
	justification	String																		
ipAddress	String																			
Operation	<table><tr><th>Name</th><th>Return Type</th></tr></table>		Name	Return Type																
Name	Return Type																			



DOCUMENT IDENTIFICATION

SYSTEM NAME	ITEM NUMBER	FORMAT	VERSION	PAGE
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	recordEvent() searchLogs() exportLogs()	Boolean Void Boolean
--	---	----------------------------

3.8.15 CSC SADashboard (TFMS-15-100-PY-CSC-15)

This CSC of two CSUs which are UI and User.

3.8.15.1 CSU UI (TFMS-15-100-PY-CSU-15.1)

The table below shows the design description of CSU UI.

Table 32: CSU UI Design Description

Type	Boundary											
Description	Represents the user interface component for the System Administrator Dashboard.											
Attributes	<table><tr><th>Name</th><th>Type</th></tr><tr><td>dashboardLayout</td><td>String</td></tr></table>		Name	Type	dashboardLayout	String						
Name	Type											
dashboardLayout	String											
Operation	<table><tr><th>Name</th><th>Return Type</th></tr><tr><td>displayDashboard()</td><td>Void</td></tr><tr><td>manageStaff()</td><td>Void</td></tr><tr><td>manageTaskForce()</td><td>Void</td></tr><tr><td>systemConfiguration()</td><td>Void</td></tr></table>		Name	Return Type	displayDashboard()	Void	manageStaff()	Void	manageTaskForce()	Void	systemConfiguration()	Void
Name	Return Type											
displayDashboard()	Void											
manageStaff()	Void											
manageTaskForce()	Void											
systemConfiguration()	Void											

3.8.15.2 CSU User (TFMS-02-100-PY-CSU-15.2)

The table below shows the design description of CSU User.

Table 33: CSU User Design Description

Type	Entity	
Description	Represents a user in the system with authentication and profile System Administrator capabilities.	
Attributes		
	Name	Type
	staffID	String
	staffName	String



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	userName password department grade category isActive	String String String String String Boolean
Operation		

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4 DETAILED DESIGN

This chapter describes the detailed design of the Task Force Management System (TFMS), focusing on the logical database structure, core business services, state transitions, security, error handling, and performance considerations. The level of detail is sufficient for implementation while remaining concise and readable.

4.1 Database Schema and Entity Overview

TFMS uses a **relational database** (PostgreSQL) with 11 main entities. This section summarizes the conceptual structure, responsibilities, and key constraints of each entity.

4.1.1 Core Entities and Responsibilities

The following table summarizes the primary responsibility of each core entity.

Table 34: Core Entities and Responsibilities

Entity	Key Responsibility
User	Staff accounts, login credentials, account status and security flags
Role	System roles and high-level permissions
UserRole	Many-to-many associations between User and Role
Department	Academic departments/faculties
TaskForce	Departmental task force definitions and lifecycle
TaskForceMembership	Assignment of staff to task forces with weightage and membership status
ApprovalRequest	HOD submissions to PSM and associated approval decisions
Report	Generated report metadata (type, date, format, filters)
WorkloadSettings	Singleton configuration for workload thresholds and selected security limits
AuditLog	Immutable audit trail of sensitive and administrative actions



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4.1.2 Key Attributes and Constraints Summary

This section highlights the most important attributes and business constraints for each major entity.

User

- Primary key: `StaffID` (e.g. `STF-0001`).
- Unique fields: `userName`, `email`.
- Security attributes: `isActive`, `must_change_password`, `failed_login_attempts`, `account_locked`.
- Organizational context: `department_id` references `Department`.

Role

- Primary key: `role_id`.
- Unique fields: `code`, `name`.
- Logical roles: `ADMIN`, `HOD`, `PSM`, `DEAN`, `LECTURER`.

UserRole

- Primary key: `user_role_id`.
- Many-to-many between `User` and `Role`.
- Composite uniqueness on (`StaffID`, `role_id`).
- `isPrimary` flag identifies the user's main role.

Department

- Primary key: `department_id`.
- Unique code per department.
- Referenced by `User` and `TaskForce`.

TaskForce

- Primary key: `taskForceId`.
- Key fields: `name`, `department_id`, `defaultWeightage`, `startDate`, `endDate`, `status`.
- Business constraint: `startDate ≤ endDate`.



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- References `Department` (owner), `User` (`created_by` , `submitted_by`).

TaskForceMembership

- Primary key: `membershipId` .
- Links `User` and `TaskForce` with `weightage` , `roleInTaskForce` , `status` , and optional `justification` .
- Composite uniqueness: (`taskForceId` , `StaffID`) .
- Constraint: $0 \leq \text{weightage} \leq 100$.

ApprovalRequest

- Primary key: `approvalId` .
- Associates a submission with a specific `TaskForce` .
- Main fields: `taskForceId` , `submittedBy` , `submissionDate` , `status` , `approvedBy` , `decisionDate` , `remarks` , `rejection_reason` .

WorkloadSettings (Singleton)

- Enforces global workload thresholds and selected security parameters.
- Only a single record exists.
- Key fields: `globalMinWeightage` , `globalMaxWeightage` , `max_failed_login_attempts` , `password_expiry_days` .

AuditLog

- Primary key: `logId` .
- Records action type, actor, affected entity, timestamp, IP address, and optional justification.
- Insert-only; no update or delete operations are allowed on this table by normal application logic.

4.1.3 Conceptual Entity Relationships

- User – Department:** Many users belong to one department.
- User – Role – UserRole:** Many-to-many between User and Role via UserRole.



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- **TaskForce – Department:** Many task forces belong to one department.
- **TaskForce – TaskForceMembership – User:** Many-to-many between TaskForce and User via TaskForceMembership.
- **TaskForce – ApprovalRequest:** One task force can have multiple approval requests across different submission cycles.
- **WorkloadSettings:** Global configuration table read by service-layer logic (no direct foreign key usage by other entities).
- **AuditLog – All Entities:** Generic reference via `entityName` and `entityId`, plus `actorStaffID`.

4.2 Service Layer and Business Logic Overview

TFMS uses a Django-style Model-View-Template (MVT) architecture. Core business rules are implemented in a logical controller/service layer, referred to here as **TFMSController**. This section presents the main operations and their essential algorithms.

4.2.1 Core Service Methods

The principal service methods are as follows:

- `authenticateUser(username, password) → UserSession`
- `manageTaskForce(operation, taskForceData) → TaskForce`
- `manageMembership(operation, membershipData) → TaskForceMembership`
- `calculateWorkload(staffId) → WorkloadStatus`
- `processApproval(approvalId, decision, remarks) → ApprovalRequest`

Only essential pseudocode is presented; low-level implementation details are omitted to keep the design document concise.

4.2.2 Workload Calculation - Essential Logic

```

FUNCTION calculateWorkload(staffID) -> WorkloadStatus
  memberships = SELECT weightage,
  status, taskForceId,
  name FROM TaskForceMembership JOIN TaskForce WHERE StaffID =
  staffID AND status IN('ACTIVE', 'PENDING')

  total = SUM(m.weightage FOR m IN memberships)

  settings = SELECT globalMinWeightage,
```



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```

globalMaxWeightage FROM WorkloadSettings WHERE setting_id =
1

IF total < settings.globalMinWeightage THEN status =
"UNDERLOADED" ELSE IF total
<= settings.globalMaxWeightage THEN status =
"BALANCED" ELSE status = "OVERLOADED" END IF

RETURN{
  StaffID : staffID,
  totalWeightage : total,
  status : status,
  globalMin : settings.globalMinWeightage,
  globalMax : settings.globalMaxWeightage,
  memberships : memberships
} END FUNCTION

```

4.2.3 Membership Management - Essential Logic

```

FUNCTION addMember(taskForceId, staffID, weightage, role, justification) ->
TaskForceMembership
  ASSERT 0 ≤ weightage ≤ 100
  ASSERT TaskForce(taskForceId) exists AND status NOT IN ('REJECTED',
'INACTIVE')
  ASSERT User(staffID) exists AND isActive = TRUE
  ASSERT no existing membership for (taskForceId, staffID)

  current = currentWorkload(staffID) // sum of ACTIVE + PENDING
  resultant = current + weightage

  settings = getWorkloadSettings()

  IF resultant > settings.globalMaxWeightage AND (justification is NULL or
empty) THEN
    RAISE OverloadedException("Exceeds maximum workload without
justification")
  END IF

  CREATE TaskForceMembership WITH
    taskForceId = taskForceId,
    StaffID = staffID,
    weightage = weightage,
    roleInTaskForce = role,
    justification = justification,
    status = 'PENDING'

  Log ADD_MEMBERSHIP in AuditLog

  RETURN created membership

```



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END FUNCTION

4.2.4 Approval Processing - Essential Logic

```

FUNCTION processApproval(approvalId, decision, remarks) ->
ApprovalRequest
  ASSERT currentUser has role PSM or ADMIN

  approval = SELECT * FROM ApprovalRequest WHERE approvalId =
approvalId
  ASSERT approval.status = 'SUBMITTED'

  IF decision = 'APPROVED' THEN
    pendingMembers = SELECT * FROM TaskForceMembership
                      WHERE taskForceId = approval.taskForceId
                      AND status = 'PENDING'

    settings = getWorkloadSettings()

    FOR EACH m IN pendingMembers DO
      IF m.weightage > settings.globalMaxWeightage AND (m.justification is
NULL or empty) THEN
        RAISE BusinessRuleViolation("Overloaded member without
justification")
      END IF
    END FOR

    UPDATE ApprovalRequest
      SET status = 'APPROVED', approvedBy = currentUser.StaffID,
        decisionDate = NOW(), remarks = remarks
      WHERE approvalId = approvalId

    UPDATE TaskForce
      SET status = 'APPROVED'
      WHERE taskForceId = approval.taskForceId

    UPDATE TaskForceMembership
      SET status = 'ACTIVE'
      WHERE taskForceId = approval.taskForceId
      AND status = 'PENDING'

  ELSE IF decision = 'REJECTED' THEN
    ASSERT remarks not empty AND LENGTH(TRIM(remarks)) ≥ 10

    UPDATE ApprovalRequest
      SET status = 'REJECTED', approvedBy = currentUser.StaffID,
        decisionDate = NOW(), rejection_reason = remarks
      WHERE approvalId = approvalId

```



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```

UPDATE TaskForce
  SET status = 'REJECTED'
  WHERE taskForceId = approval.taskForceId

// Memberships remain PENDING for HOD to revise and resubmit

ELSE
  RAISE InvalidDecision("Decision must be APPROVED or REJECTED")
END IF

Log APPROVE_REQUEST or REJECT_REQUEST in AuditLog

RETURN updated ApprovalRequest
END FUNCTION

```

4.3 State Transition Specifications

This section defines the main lifecycle states and transitions for TaskForce, TaskForceMembership, and ApprovalRequest in a concise, implementation-oriented form.

4.3.1 TaskForce Status Lifecycle

The following pseudocode summarizes the allowed TaskForce state transitions.

STATES: ACTIVE, SUBMITTED, APPROVED, REJECTED, INACTIVE

INITIAL:

On creation → status = ACTIVE

EVENT submitForApproval (by HOD)

PRE: status = ACTIVE

ACTION:

status = SUBMITTED

create ApprovalRequest(status = 'SUBMITTED')

EVENT approve (by PSM)

PRE: status = SUBMITTED

ACTION:

status = APPROVED

ApprovalRequest.status = 'APPROVED'

all PENDING memberships for this TaskForce → ACTIVE

EVENT reject (by PSM)

PRE: status = SUBMITTED

ACTION:

status = REJECTED



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ApprovalRequest.status = 'REJECTED'
memberships remain PENDING for revision

EVENT resubmit (by HOD after REJECTED)

PRE: status = REJECTED

ACTION:

create NEW ApprovalRequest(status = 'SUBMITTED')
status = SUBMITTED

EVENT deactivateOrEnd

PRE: status IN ('APPROVED', 'ACTIVE', 'REJECTED')

ACTION:

status = INACTIVE
if previously APPROVED: ACTIVE memberships → COMPLETED

4.3.2 TaskForceMembership Status Lifecycle

The following pseudocode summarizes TaskForceMembership state transitions.

STATES: PENDING, ACTIVE, COMPLETED, DROPPED/REMOVED

INITIAL:

On addMember → status = PENDING

EVENT parentApproved

PRE: membership.status = PENDING

parent TaskForce.status = APPROVED

ACTION:

membership.status = ACTIVE

EVENT taskForceEnds

PRE: membership.status = ACTIVE

parent TaskForce.status = INACTIVE

ACTION:

membership.status = COMPLETED

EVENT manualRemove (by HOD or ADMIN)

PRE: membership.status IN ('PENDING', 'ACTIVE')

ACTION:

membership.status = DROPPED/REMOVED

EVENT editBeforeApproval

PRE: parent TaskForce.status NOT IN ('SUBMITTED', 'APPROVED')

ACTION:

update fields; membership.status remains PENDING



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4.3.3 ApprovalRequest Status Lifecycle

The table below summarizes ApprovalRequest state transitions.

STATES: SUBMITTED, APPROVED, REJECTED, WITHDRAWN

INITIAL:

On HOD submit → status = SUBMITTED

EVENT approve (by PSM)

PRE: status = SUBMITTED

ACTION: status = APPROVED (terminal)

EVENT reject (by PSM)

PRE: status = SUBMITTED

ACTION: status = REJECTED (terminal)

EVENT withdraw (by HOD)

PRE: status = SUBMITTED

ACTION: status = WITHDRAWN (terminal)

4.4 Security and Access Control Overview

4.4.1 Authentication and Login

- Credentials: `userName` and password.
- Passwords are stored as **PBKDF2-SHA256 hashes** with per-user salts.
- After three consecutive failed login attempts, `account_locked` is set to TRUE and the account must be unlocked by an ADMIN.
- On successful login, `failed_login_attempts` is reset to 0 and a secure session is created using HTTP-only, Secure cookies with a 30-minute idle timeout.

4.4.2 Role Based Access Control (RBAC)

- **ADMIN:** Full system administration, user management, configuration, and audit access.
- **HOD:** Manages task forces within own department (create, edit, submit for approval, manage members).
- **PSM:** Reviews and approves/rejects ApprovalRequests; views workload and departmental reports.
- **DEAN:** Views faculty-level analytics and summary reports.
- **LECTURER:** Views own workload and participation only.



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Access to each business operation is guarded by both:

- The user's assigned roles, and
- Contextual checks (e.g. department ownership, entity status).

4.4.3 Audit Logging

TFMS records all security-sensitive and business-critical actions in **AuditLog**, including but not limited to:

- Authentication events: login, logout, failed login attempts, account lock/unlock.
- TaskForce lifecycle events: create, update, submit, approve, reject, deactivate.
- Membership changes: add, update, remove.
- Configuration changes: modifications to WorkloadSettings and security parameters.

AuditLog entries are immutable, retained for at least seven years, and visible only to authorized administrative roles.

4.5 Error Handling Logical View

4.5.1 Main Exception Categories

- **Authentication and Authorization**
 1. Invalid credentials
 2. Account locked
 3. Unauthorized access to operations or entities
- **Validation and Business Rules**
 1. Invalid field values or missing mandatory data
 2. Overloaded workload assignments without justification
 3. Invalid state transitions (e.g. approving a non-SUBMITTED request)
- **Data Access**
 1. Entity not found
 2. Duplicate entity (e.g. duplicate membership in the same TaskForce)

The presentation layer maps these technical exceptions to clear, user-friendly error messages and, where possible, suggests corrective actions.

4.6 Performance Considerations

Key performance considerations include:

- Appropriate indices on frequently filtered columns such as:



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1. User.userName , User.isActive , User.department_id
2. TaskForce.status , TaskForce.department_id , TaskForce.startDate , TaskForce.endDate
3. TaskForceMembership.StaffID , TaskForceMembership.status , TaskForceMembership.taskForceId
4. ApprovalRequest.status , ApprovalRequest.submissionDate
5. AuditLog.timeStamp , AuditLog.actorStaffID

- Expected performance under normal load with indexing:
 1. Login validation: under 5 ms.
 2. Workload calculation per staff: under 10 ms.
 3. Departmental approval dashboard queries: under 20 ms.
- Lightweight caching:
 1. WorkloadSettings cached in memory (changes are infrequent).
 2. Role information cached within user session for the logged-in user.

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5 CSCI DATA

This chapter describes the key data elements, validation rules, lifecycle, integrity constraints, and retention requirements for TFMS. The focus is on clarity and traceability between business rules and the underlying data.

5.1 Data Dictionary Summary

This section presents the principal data elements for the main entities in tabular form.

5.1.1 User Entity Data Elements

The following table summarizes the core data elements maintained for each system user.

Table 35: User Entity Data Elements

Element	Type / Size	Constraints / Range	Description
staffID	VARCHAR(10)	PK, unique, format e.g. STF-0001	Staff identifier; primary key
userName	VARCHAR(50)	Unique, required, 5–50 chars	Login ID
password_hash	VARCHAR(255)	Required; PBKDF2-SHA256 hash only	Hashed password (never plaintext)
email	VARCHAR(100)	Unique, optional, valid email format	Notification and password recovery email
department_id	INT	FK to Department, required	Staff department
grade	VARCHAR(30)	Optional, from predefined list	Academic rank (e.g. Professor, Lecturer)
category	VARCHAR(30)	Optional, from predefined list	Staff category (Academic, Support, etc.)
isActive	BOOLEAN	Default TRUE	Can the user log in
must_change_password	BOOLEAN	Default TRUE for new users	Forces password change at first login



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Element	Type / Size	Constraints / Range	Description
failed_login_attempts	INT	0–3	Counter for failed logins
account_locked	BOOLEAN	Default FALSE; set TRUE after 3 failures	Account lock flag
created_at	TIMESTAMP (UTC)	Auto-set	Record creation time
updated_at	TIMESTAMP (UTC)	Auto-updated	Last modification time

5.1.2 TaskForce Entity Data Elements

The following table lists the main data elements for task forces.

Table 36: Taskforce Entity Data Elements

Element	Type / Size	Constraints / Range	Description
taskForceId	INT	PK, auto-increment	Task force identifier
name	VARCHAR(100)	Required, 5–100 chars	Task force name
description	TEXT	Optional	Purpose and scope
department_id	INT	FK to Department, required	Owning department
defaultWeightage	INT	0–100, default 0	Default suggested weightage per member
startDate	DATE	Required; $\text{startDate} \leq \text{endDate}$	Task force start date
endDate	DATE	Required; $\text{endDate} \geq \text{startDate}$	Task force end date
status	VARCHAR(20)	Enum: ACTIVE, SUBMITTED, APPROVED, REJECTED, INACTIVE	Current lifecycle state
created_by	VARCHAR(10)	Optional FK to User	Creator of the task force
submitted_by	VARCHAR(10)	Optional FK to User	HOD who submitted for approval
submitted_at	TIMESTAMP (UTC)	NULL until first submission	Submission timestamp
created_at	TIMESTAMP (UTC)	Auto-set	Record creation time



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Element	Type / Size	Constraints / Range	Description
updated_at	TIMESTAMP (UTC)	Auto-updated	Last modification time

5.1.3 TaskForceMembership Entity Data Elements

The following table captures the data elements describing staff membership in task forces.

Table 37: TaskForceMembership Entity Data Elements

Element	Type / Size	Constraints / Range	Description
membershipId	INT	PK, auto-increment	Membership identifier
taskForceId	INT	FK to TaskForce, required	Task force reference
staffID	VARCHAR(10)	FK to User, required	Member (staff) reference
roleInTaskForce	VARCHAR(50)	Optional, 5–50 chars	Role in task force (Chair, Secretary, Member)
weightage	INT	Required, 0–100	Workload contribution
startDate	DATE	Optional; if set, startDate ≤ endDate	Membership start date
endDate	DATE	Optional; if set, endDate ≥ startDate	Membership end date
justification	TEXT	Required if resultant workload exceeds max limit	Reason for assigning/overloading
status	VARCHAR(20)	Enum: PENDING, ACTIVE, COMPLETED, DROPPED/REMOVED	Membership lifecycle state
created_at	TIMESTAMP (UTC)	Auto-set	Record creation time
updated_at	TIMESTAMP (UTC)	Auto-updated	Last modification time

5.1.4 ApprovalRequest Entity Data Elements

The following table lists the fields used to track approval requests.



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Table 38: ApprovalRequest Entity Data Elements

Element	Type / Size	Constraints / Range	Description
approvalId	INT	PK, auto-increment	Approval request ID
taskForceId	INT	FK to TaskForce, required	Task force being reviewed
submittedBy	VARCHAR(10)	FK to User (HOD), required	Who submitted the request
submissionDate	TIMESTAMP (UTC)	Auto-set	When the request was submitted
status	VARCHAR(20)	Enum: SUBMITTED, APPROVED, REJECTED, WITHDRAWN	Approval lifecycle state
approvedBy	VARCHAR(10)	Optional FK to User (PSM)	Who approved/rejected
decisionDate	TIMESTAMP (UTC)	NULL until decision	When the decision was made
remarks	TEXT	Optional for APPROVED	General comments by PSM
rejection_reason	TEXT	Required if status = REJECTED	Justification for rejection
created_at	TIMESTAMP (UTC)	Auto-set	Record creation time
updated_at	TIMESTAMP (UTC)	Auto-updated	Last modification time

5.1.5 WorkloadSettings Entity Data Elements

The following table summarizes the singleton configuration parameters controlling workload and selected security policies.

Table 39: WorkloadSettings Entity Data Elements

Element	Type / Size	Constraints / Range	Description
setting_id	INT	Always 1; PK; singleton	Single configuration record
globalMinWeightage	INT	0–100; default 0	Minimum workload threshold



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globalMaxWeightage	INT	0–100; default 30; $\geq$ min	Maximum recommended workload threshold
max_failed_login_attempts	INT	1–10; default 3	Login attempts before account lockout
password_expiry_days	INT	0+; default 0 (disabled)	Password expiry interval in days (if enabled)
last_modified	TIMESTAMP	Auto-updated	When settings were last changed
modified_by	VARCHAR(10)	Optional FK to User	Who last modified the settings

5.1.6 AuditLog Entity Data Elements

The following table describes the structure of the audit trail maintained by TFMS.

Table 40: AuditLog Entity Data Elements

Element	Type / Size	Constraints / Range	Description
logId	INT	PK, auto-increment	Audit log entry ID
actionType	VARCHAR(50)	Required; from defined action list	Type of action (LOGIN, APPROVE_REQUEST, etc.)
entityName	VARCHAR(100)	Optional	Name of affected entity (e.g. TaskForce)
entityId	INT	Optional	Primary key of affected entity
actorStaffID	VARCHAR(10)	Optional FK to User	Who performed the action
actionDetails	TEXT	Optional; often JSON	Additional context (old/new values, etc.)
ipAddress	VARCHAR(45)	Optional	Client IP address
timeStamp	TIMESTAMP (UTC)	Auto-set; immutable	When the action occurred
justification	TEXT	Required for certain actions	Reason for sensitive changes or overrides

5.2 Core Validation Rules

This section summarizes key validation rules enforced either at the database level or in the service layer. The following table lists the main validation rules by area.

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Table 41: Core Validation Rules

Area	Rule
Weightage	$0 \leq \text{weightage} \leq 100$ for every TaskForceMembership.
Workload limit	If $\text{currentWorkload} + \text{newWeightage} > \text{globalMaxWeightage}$, justification is mandatory.
TaskForce dates	<code>startDate</code> and <code>endDate</code> required; must satisfy $\text{startDate} \leq \text{endDate}$.
Membership dates	If both set, $\text{membership.startDate} \leq \text{membership.endDate}$.
Approval status	$\text{status} \in \{\text{SUBMITTED}, \text{APPROVED}, \text{REJECTED}, \text{WITHDRAWN}\}$ only.
Rejection reason	If $\text{status} = \text{REJECTED}$, <code>rejection_reason</code> required and meaningfully completed.
User uniqueness	<code>StaffID</code> , <code>userName</code> , and <code>email</code> must each be unique.
Membership unique	Only one membership per $(\text{taskForceId}, \text{StaffID})$ pair.

5.3 Data Lifecycle and State Transitions

This section provides a concise view of how entity states evolve over time.

5.3.1 TaskForce Status Transitions

The following table describes the allowed TaskForce state transitions.

Table 42: TaskForce Status Transitions

From State	Event / Trigger	To State	Notes
(none)	Create TaskForce	ACTIVE	Initial creation by ADMIN/HOD
ACTIVE	HOD submits for approval	SUBMITTED	ApprovalRequest created
SUBMITTED	PSM approves	APPROVED	All PENDING memberships become ACTIVE
SUBMITTED	PSM rejects	REJECTED	HOD may edit and submit a new ApprovalRequest
REJECTED	HOD resubmits (new ApprovalRequest)	SUBMITTED	New request ID; previous request remains in history
APPROVED	End date reached or admin deactivation	INACTIVE	ACTIVE memberships move to COMPLETED



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ACTIVE	Admin deactivates	INACTIVE	Task force closed without approval
REJECTED	Admin deactivates	INACTIVE	Rejected task force abandoned

5.3.2 TaskForceMembership Status Transitions

The following table outlines how membership records progress through their lifecycle.

Table 43: TaskForceMembership Status Transitions

From State	Event / Trigger	To State	Notes
(none)	Member added to TaskForce	PENDING	Created by HOD/ADMIN
PENDING	Parent TaskForce approved by PSM	ACTIVE	Automatic transition
ACTIVE	Parent TaskForce becomes INACTIVE	COMPLETED	Membership ends with task force
PENDING, ACTIVE	Manual removal (HOD/ADMIN)	DROPPED/REMOVED	Member explicitly removed before/after approval
PENDING	Edit before submission/approval	PENDING	Fields may change; state remains PENDING

5.3.3 ApprovalRequest Status Transitions

The following table shows ApprovalRequest lifecycle transitions.

Table 44: ApprovalRequest Status Transitions

From State	Event / Trigger	To State	Notes
(none)	HOD submits TF	SUBMITTED	Request created and linked to a TaskForce
SUBMITTED	PSM approves	APPROVED	Terminal; associated TaskForce set to APPROVED
SUBMITTED	PSM rejects	REJECTED	Terminal; TaskForce set to REJECTED



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SUBMITTED	HOD withdraws	WITHDRAWN	Terminal; request cancelled by HOD
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5.4 Data Quality and Integrity

The following table summarizes the key mechanisms used to maintain data quality and referential integrity.

Table 45: Data Quality and Integrity

Aspect	Key Rules / Mechanisms
Primary keys	All main entities have primary keys (e.g. StaffID, taskForceId, membershipId).
Foreign keys	Enforced for relationships between User, Department, TaskForce, Membership, Approval.
Uniqueness	Enforced for StaffID, userName, email, role code/name, department code, membership pair.
Check constraints	Enforce numeric ranges and valid status values where applicable.
Referential integrity	Orphan records prevented through CASCADE or SET NULL behaviour as appropriate.
Auditability	All critical lifecycle changes and security-sensitive operations logged in AuditLog.

5.5 Retention and Security Summary

The following table provides a high-level view of retention and security requirements for each major data category.

Table 46: Retention and Security Summary

Data Category	Retention / Handling	Notes
TaskForce & Membership	At least 7 years after completion or inactivation	Supports historical workload analysis
ApprovalRequest	At least 7 years	Supports decision and audit trail
AuditLog	At least 7 years	Compliance and forensic investigations
User accounts	Active: indefinite; inactive: retained per policy	Certain fields may be anonymized if needed



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Passwords	Stored only as PBKDF2-SHA256 hashes; never in plaintext	Used solely for verification
Transport security	All TFMS access over HTTPS (TLS 1.2 or higher)	Protects credentials and personal data
Access control	RBAC enforced; sensitive data (AuditLog, settings) ADMIN-only	All access is logged and monitored

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6 NOTES

Abbreviation used:

- **CDRL** Context Data Requirement List
- **CSC** Computer Software Component
- **CSCI** Computer Software Configuration Item
- **CSU** Computer Software Unit
- **DOD** Department of Defense
- **FAI** Faculty of Artificial Intelligence
- **IRS** Interface Requirement Specification
- **MIL** Military
- **MVT** Model View Template
- **OBA** On Board Automobile
- **OOAD** Object Oriented Analysis Design
- **SDP** Software Development Plan
- **SRS** Software Requirement Specification
- **SOW** Statement of Work
- **TFMS** Taskforce Management System
- **UML** Unified Modeling Language
- **UTM** Universiti Teknologi Malaysia

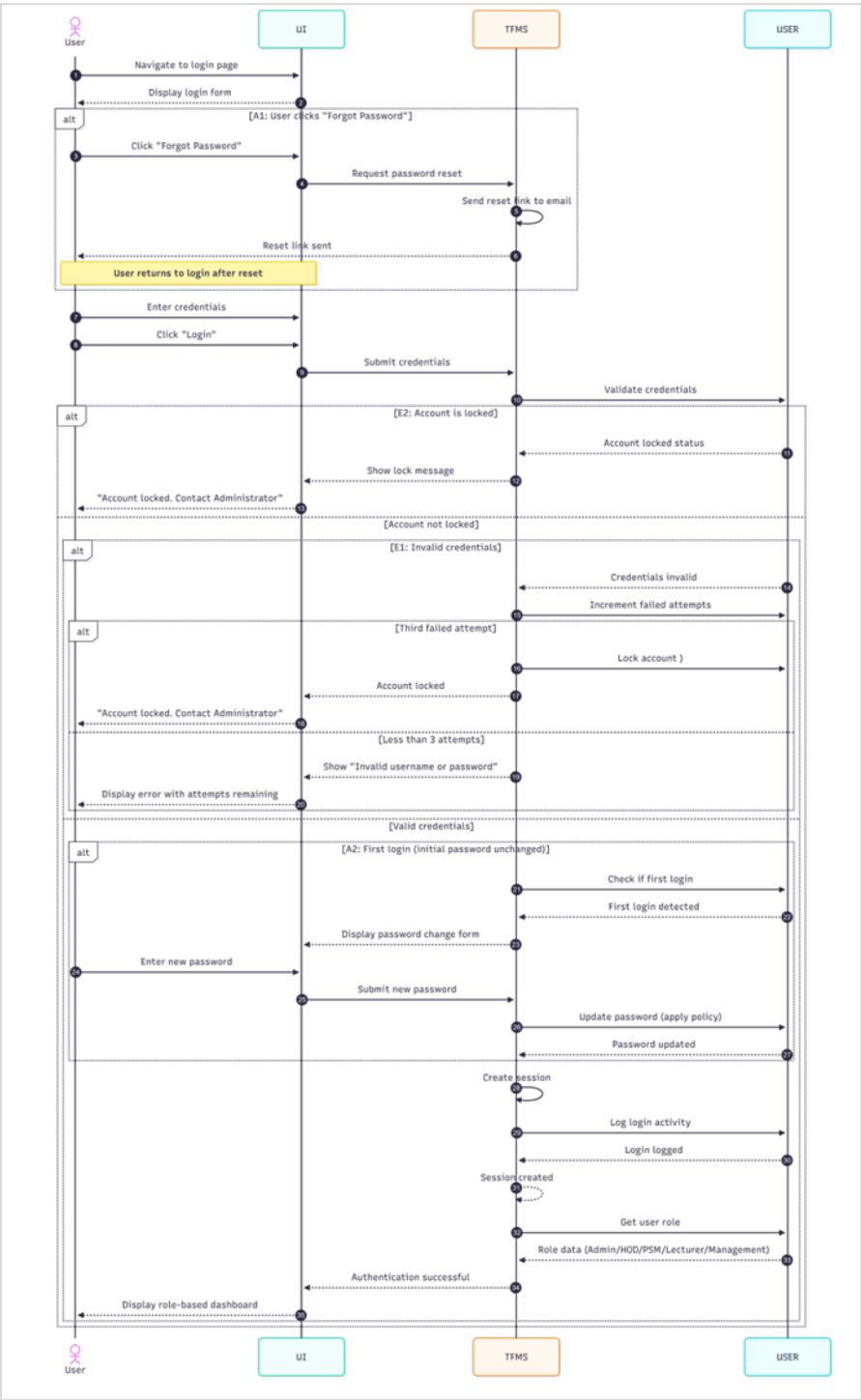


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7 ANNEX 1

Appendix 1: Sequence Diagram - User Authentication and Login



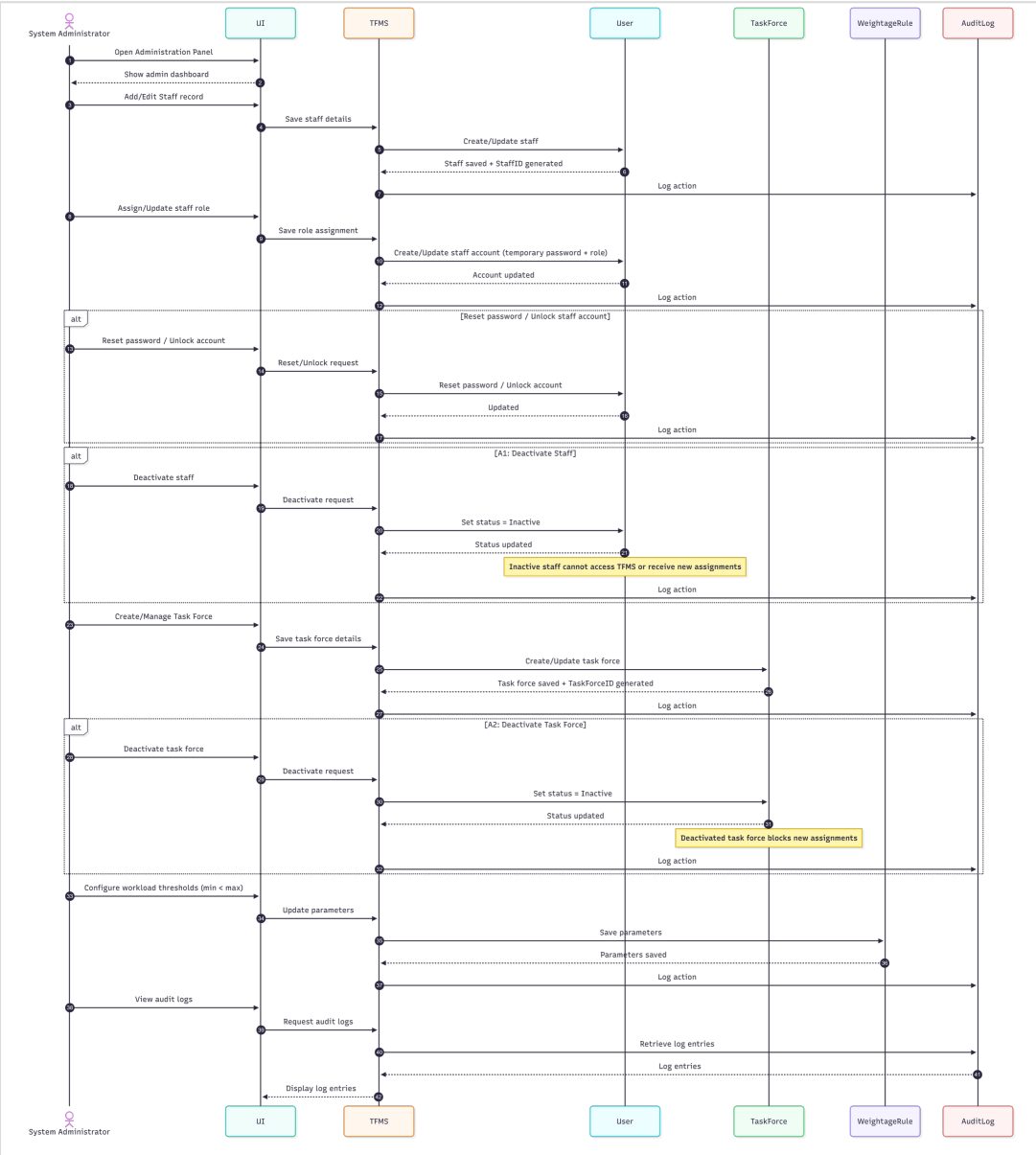
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Appendix 2: Sequence Diagram - Configure TFMS Parameters and Master Data Management



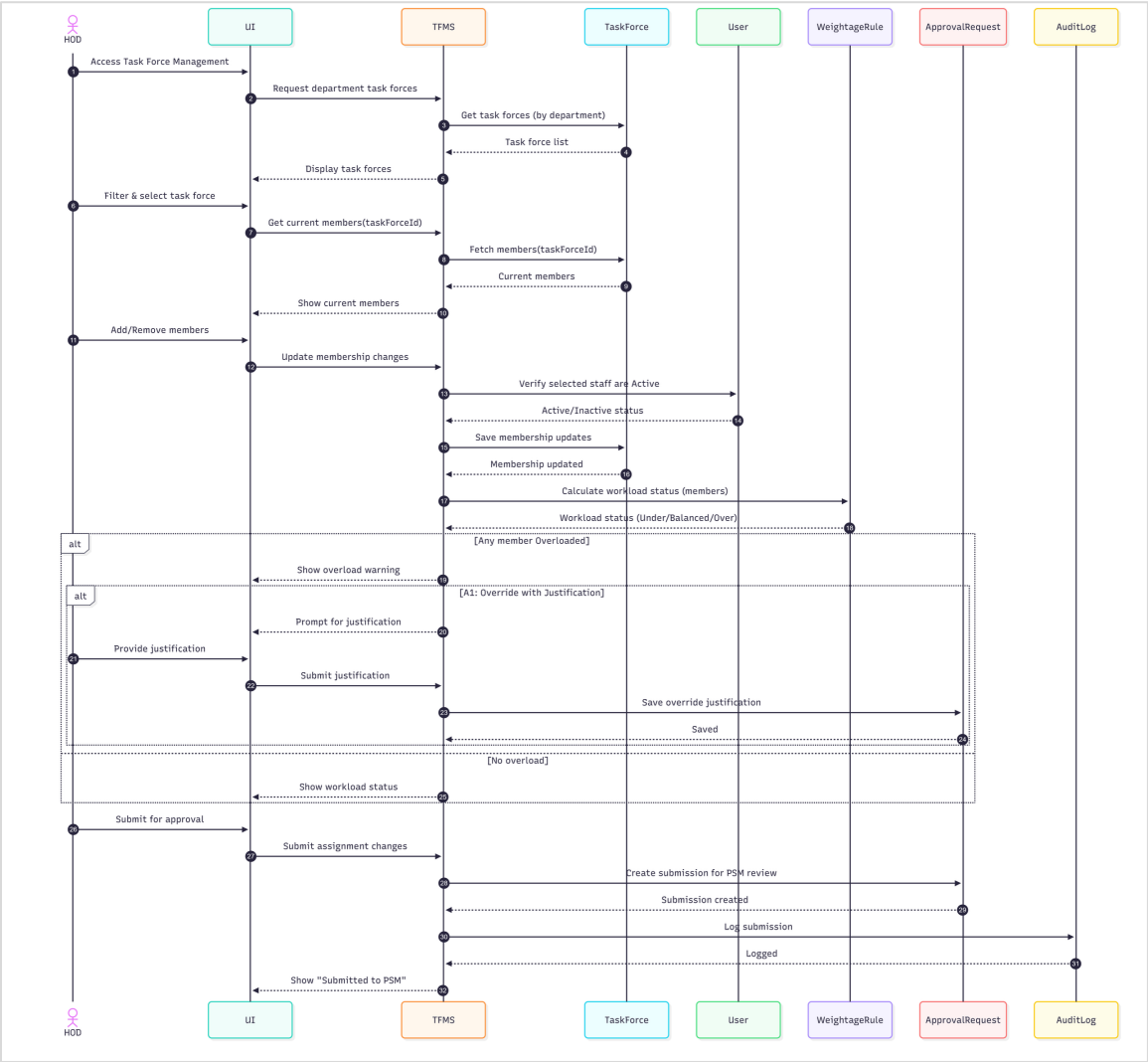
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Appendix 3: Sequence Diagram - Manage Task Force Membership and Workload Distribution



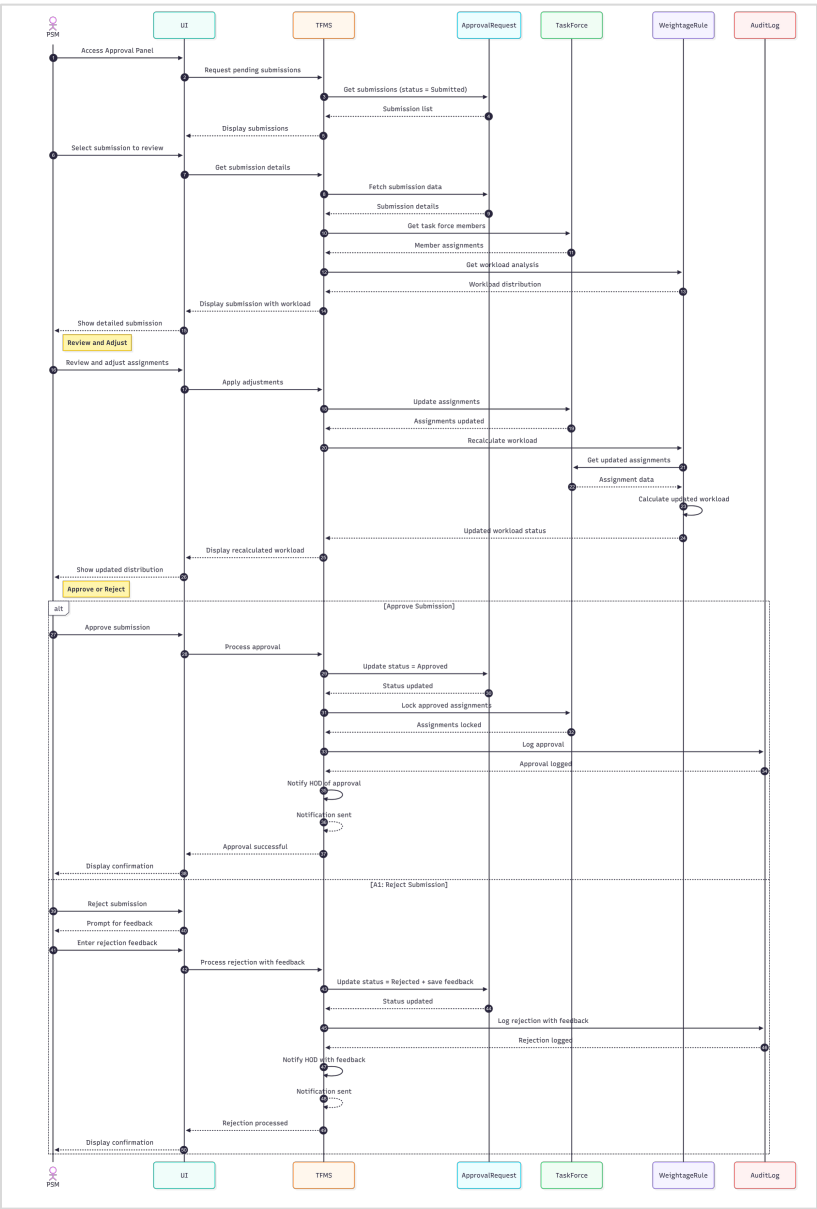
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Appendix 4: Sequence Diagram - Review and Approve Task Force Assignments



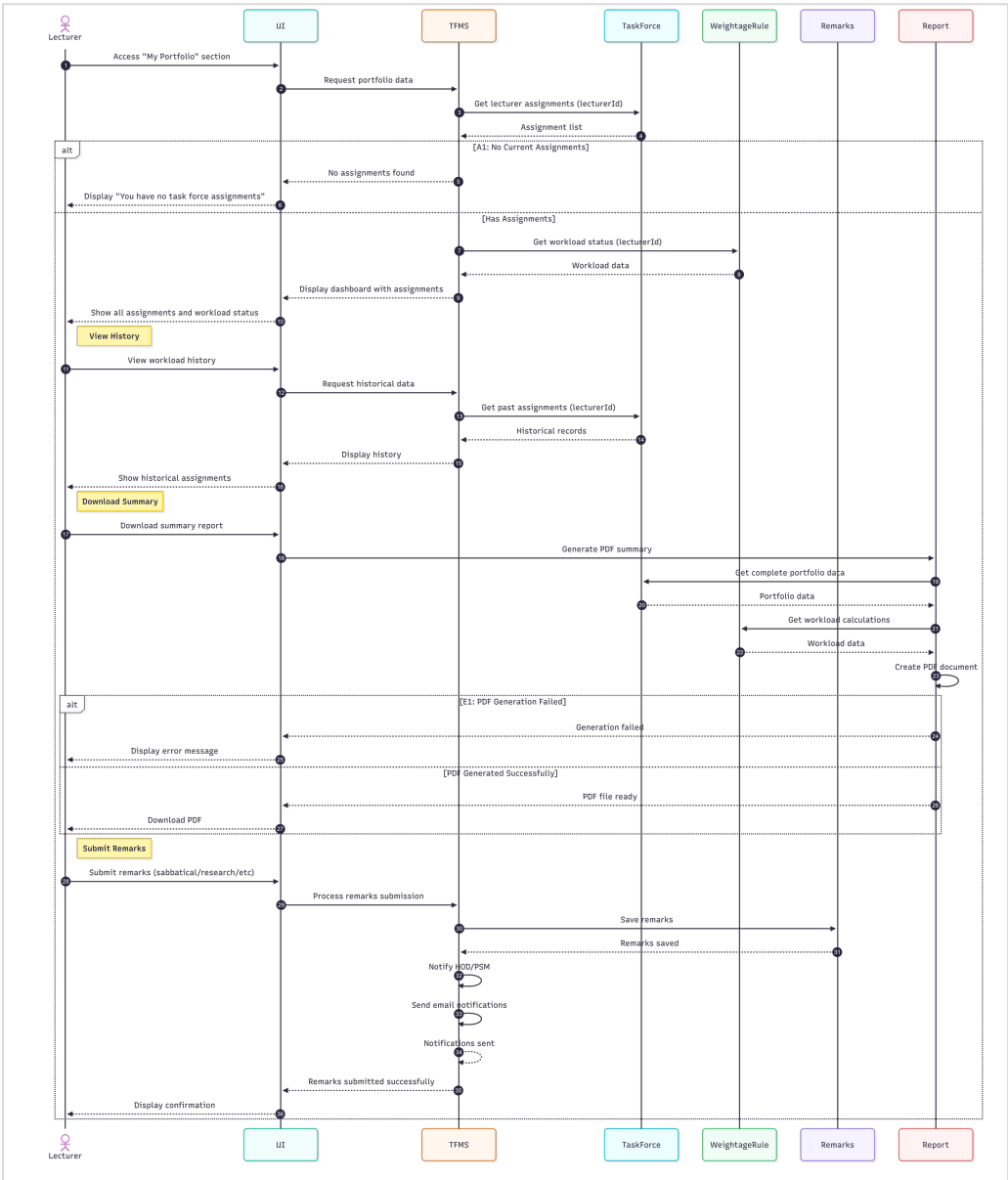
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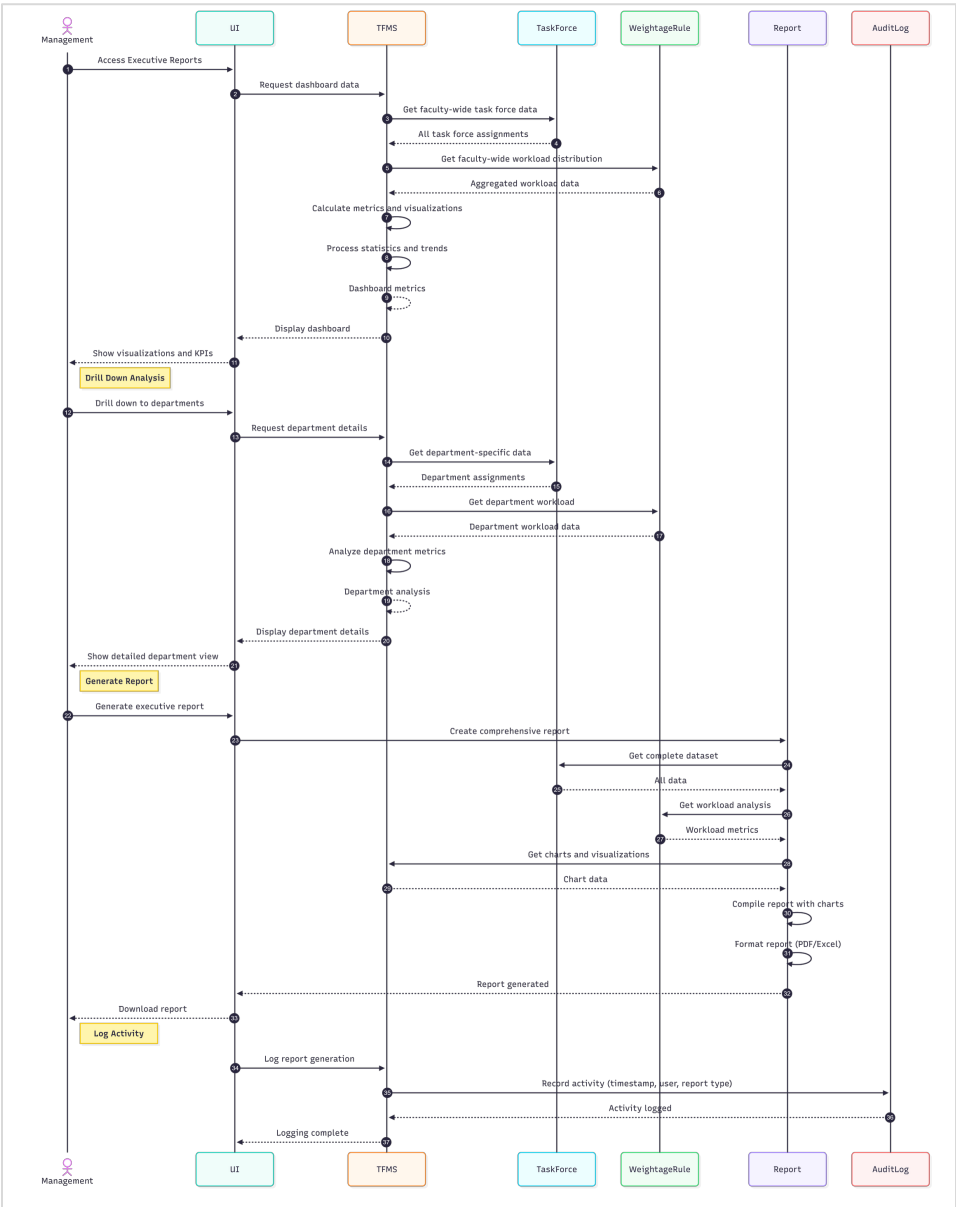
Appendix 5: Sequence Diagram - Access Personal Task Force Portfolio



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Appendix 6: Sequence Diagram - Generate Executive Reports and Analytics



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